Patient-Wise Splitting Degrades Into Random Splitting as Participant Identifiers Fail: A Controlled Measurement in Alzheimer's Disease MRI

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Abstract

Patient-wise splitting is the standard remedy for leakage in medical imaging, and its guarantee is usually credited to the splitting procedure. It is not a property of the procedure. It is a property of the participant identifier the procedure is given, and identifiers are deleted, fragmented and merged by redistribution. We measure that dependence across its full range on ADNI1 (**220** cognitively normal and AD participants, **1,123** scans, one 2D coronal-centre slice per scan). With identifiers complete, subject-wise splitting reports an area under the receiver operating characteristic curve (AUROC) of **0.771**; with every identifier removed the same unchanged procedure reports **0.946**, against **0.947** for random image-level splitting, an inflation of **+0.176** accrued without any error being raised. A provenance-aware grouping adds nothing while identifiers are intact, which is the design target, and declines optimism in **7** of **8** degraded conditions (Wilcoxon $p = 0.008$). Permuting diagnosis across participants removes the disease signal, yet random splitting still reaches **0.878** AUROC while both patient-aware protocols sit at chance, identifying patient recall as the mechanism. One instance is observed rather than imposed: a widely used public 2D AD benchmark is, by content matching, a redistribution of **200** OASIS-1 participants, and grouping by the only identifier it supplies still leaves **70%** of test images sharing a

participant with training. Manifests, audit code and the rebuild pipeline are released under Apache-2.0.

Keywords: Patient-identity leakage, Evaluation protocol, Alzheimer's disease, Structural MRI, Data provenance, Reproducibility

1 Introduction

Alzheimer's disease (AD) is the leading cause of dementia worldwide [1] and, under the NIA-AA diagnostic framework [2], a major target for medical-imaging deep learning. The modelling literature is mature: 3D residual networks [3], hierarchical fully-convolutional architectures that jointly localise atrophy and classify [4], and interpretability-aware frameworks validated against neuropathology [5]. Reported benchmark accuracy on structural MRI is routinely near-perfect: a recent scoping review of 44 AD deep-learning studies finds a 19-point accuracy gradient between studies judged at low and at high risk of leakage [6]. These reports, however, rarely survive patient-independent evaluations. One important cause is *patient-identity leakage* across train and test partitions. It arises in AD MRI through four channels (Figure 1): slices of one 3D volume distributed across partitions; different visits from one longitudinal record; near-duplicate or augmentation-derived images on opposite sides of a split; and rescans of the same session. Because AD MRI research routinely uses longitudinal cohorts with multiple visits per patient (ADNI: median 5 scans/subject) as well as slice-based 2D representations, each of these channels is a live risk in practice. Under any of them, the trained model recognises images from patients it has already analysed rather than generalising to new patients [6–8].

That leakage inflates reported performance is established. The evidence includes slice-wise against subject-wise splitting on four cohorts [8], a review recommending subject-level partitioning [7], five leakage forms manipulated in connectome models [9], and a 19-point accuracy gradient between studies at low and high risk of leakage in a scoping review of 44 [6].

Why *patient identity* is a particularly important shortcut channel in longitudinal structural MRI is equally well established. Shortcut learning [10] exploits stable but clinically irrelevant cues correlated with the label, as with hospital-system markers in chest radiographs [11], non-lung cues in COVID classifiers [12] and hidden subgroup failures [13]. Structural MRI is unusually exposed to the identity version: face recognition re-identifies anonymised scans [14], *BrainPrint* is a stable per-subject fingerprint [15], learned representations support cross-cohort subject re-identification [16], and even brain-age prediction carries subject-specific structural signal [17]. A model evaluated on subject-leaky data can therefore appear to recognise disease while partly recognising the patient, a protocol-driven reliability gap also documented in COVID imaging [18], in medical-imaging machine learning generally [19], and as a cross-disciplinary leakage taxonomy [20].

The remedy is likewise available: subject-aware splitting ships in general-purpose libraries (`GroupKFold` [21]) and medical-imaging frameworks (MONAI [22]), a similarity-aware splitter has been proposed [23], and CLAIM [24, 25], TRIPOD+AI [26] and PROBAST [27] all require subject-level splitting for a low risk-of-bias verdict. Protocol B groups by `subject_id`,

the key those guidelines prescribe, though it is this paper's own participant-wise splitter rather than `GroupKFold` itself (§3.5.1).

What these studies leave unmeasured is how evaluation optimism changes with leakage magnitude, and how provenance degradation determines the leakage that survives a patient-aware split. Two quantities govern how much optimism a benchmark reports, and neither has been characterised. The first is how much optimism a unit of leakage buys. Rosenblatt et al. come closest, reporting performance deltas at three discrete levels of subject repetition (5, 10 and 20%) in connectome models, but they fit no function across those levels and state no per-unit rate. The second has not been asked at all: what determines how much leakage a given benchmark admits in the first place. That is a question about data provenance. A splitting protocol can only keep a patient's images together if the identifiers it depends on are present and correct; curated research cohorts supply them, and the redistributed benchmarks the field actually trains on frequently do not.

In this work we measure both quantities on a single cohort, using one experimental template for each: hold everything fixed, sweep one controlled quantity, and read off the curve.

We measure both links of that chain on a single cohort in one code state, manipulating the two quantities independently. Provenance integrity is varied by deleting participant identifiers at controlled rates, and by corrupting them in the two ways redistribution actually corrupts them: dealing one participant across several keys, and merging several under one. Test-set contamination is varied separately, by substituting a measured share of test scans with other scans of training participants, which also removes those scans from training. A participant-label permutation serves as the positive control: with the association between anatomy and diagnosis destroyed, any discrimination a protocol retains is patient recall and nothing else. One case is observed rather than imposed, in a widely used public benchmark that redistributes a research cohort under an identifier that merges and splits its participants.

This paper makes three contributions.

First, and principally, it measures the function relating evaluation validity to identifier quality, where the literature has established only its two endpoints as categories. Random splitting is known to inflate and patient-wise splitting is known to correct it; what lies between has not been measured. We show that the relation is continuous and that at its far end the correction disappears: with every participant identifier removed, subject-wise splitting reports 0.946 against 0.947 for random splitting on the same cohort, an inflation of +0.176 that the procedure signals in no way. Patient-wise splitting is therefore not a property of a splitting algorithm but of the evidence the algorithm is given, and a paper that reports having used it has reported a procedure rather than a guarantee.

Second, it calibrates what the surviving leakage is worth, through a contamination dose-response and a permutation control, the latter an established instrument [28] applied here to separate patient recall from disease signal.

Third, it releases SplitGuard-AD, an executable audit that builds a per-image leakage graph from whatever identifier evidence a cohort carries, emits a frozen component-safe partition and reports a pre-training decision, together with the manifests, component tables and rebuild pipeline behind every number reported here. We offer it as the instrument that makes the measurement above possible rather than as an improvement on participant grouping. Where

identity is intact its component layer reduces to participant grouping, as it should, and the benefit it does carry appears only once provenance is degraded. Both halves of that statement are reported as results.

The observation that random image-level splits inflate AD MRI scores is not new by itself. Wen *et al.* [7] and Yagis *et al.* [8] quantified the inflation on individual datasets. Rumala [29] showed on longitudinal brain MRI that folds which do not separate subjects inflate 3D CNN estimates, and used saliency maps to argue that the network attends to subject identity. Ghiasi *et al.* [30] established a leakage-free longitudinal dementia benchmark by withholding the label-defining clinical score and splitting group-aware, reporting the inflated and clean conditions as an on/off pair. Subject-aware splitting tools such as DataSAIL [23] are available for biomedical data. Separately, biometric work on MRI representations [14–16] has established that structural MRI leaks subject identity at re-identification accuracies far above chance. Two concurrent audits reach the far end of the axis measured here from the other direction, by finding benchmarks already there. Sifat *et al.* [31] trace subject identifiers from the slice captions of the same Kaggle redistribution studied here and report that its subject count does not match the labelled source cohort. Vangala *et al.* [32] recover patient identity for the traceable 12.6% of a public brain-tumour test split by matching to the original release, find that 95.5% of that subset shares a participant with training, and report the stronger result that removing every near-duplicate test image leaves balanced accuracy unchanged. We take both as evidence that the condition is common rather than exotic, which is the premise this paper needs and does not have to establish alone. Neither measures the path between the endpoints, and neither retrains: the quantity they report is what a contaminated benchmark yields, not how that quantity varies with the integrity of the identifier. We return to the apparent disagreement between their null and our gap in §4.2, where it resolves into a difference in what the two designs can detect.

Across all of it, leakage is treated as a binary hazard: either the split respects patients or it does not, and the remedy is to group by subject. That framing presumes what it needs, a subject identifier that is present, complete and correct, which is the presumption this paper measures. Supplementary Table S4 places the present work against that literature on the axes that separate them.

The cross-tier inflation gap is validated on two source cohorts and one redistribution of one of them; the ADNI-specific findings (subject-level aggregation, sex disparity, clinical cost, dose-response) are single-cohort findings on ADNI1 itself, and external-cohort replication (AIBL [33], UK Biobank [34], OASIS-3 [35]) is explicitly named as the immediate next step (§4.4) rather than as a contribution. We do not run head-to-head comparisons against general-purpose splitting tools or biometric re-identification models. The contribution is the measurement: that patient-independent evaluation has an untested dependency on provenance integrity, and that both links of the chain from degraded provenance to admitted leakage to evaluation optimism can be quantified experimentally. SplitGuard-AD is the audit that operationalises it, and its component layer is deliberately equivalent to participant grouping where participant identity is intact; we report that equivalence as a finding rather than defending it as a feature.

Table 1: Leakage and shortcut taxonomy for Alzheimer's MRI deep learning. Risk levels are indicative for a redistributed public 2D benchmark and reflect the typical exposure of a CN-vs-AD slice-level classifier; absolute risk in any new study depends on cohort-specific structure.

Mode	Risk	Description
Subject leakage	Critical	Same patient in train and test.
Session leakage	High	Same visit/acquisition across splits.
Longitudinal leakage	High	Same patient at different time points.
Slice leakage	High	2D slices from one volume cross splits.
Near-duplicate	Moderate	Perceptually similar derived images.
Source leakage	Moderate	Dataset/collection provenance artifact.
Site/scanner	Moderate	Scanner, field strength, protocol artifact.
Demographic shortcut	Low–Mod	Age/sex imbalance correlated with label.
Clinical-label shortcut	Low	Advanced atrophy makes task trivially easy.

2 Materials and Methods

2.1 A leakage taxonomy for slice-based AD MRI

We enumerate nine leakage and shortcut modes relevant to AD MRI deep learning. The list is intended as an operational checklist for AD MRI benchmarks rather than a closed taxonomy of every conceivable shortcut; the modes below are drawn from the prior leakage and shortcut literature [7, 8, 10, 13, 18, 20] and consolidated here into a structured form whose risk labels are calibrated to AD MRI deep-learning practice. Table 1 provides a structured overview. The new element relative to the existing leakage taxonomies is the explicit treatment of the near-duplicate (image-content) mode and the longitudinal-converter mode as first-class entries: both are characteristic of AD MRI workflows (augmented public releases, multi-visit research cohorts) and are not separately enumerated in general-purpose taxonomies such as Kapoor and Narayanan [20].

Subject, session, longitudinal, and slice leakage all violate the same clinical independence assumption: a test case should represent a new patient, not another view, visit, or derivative of a patient already seen during training. This is especially important for 2D MRI pipelines because repeated slices from one 3D scan preserve head shape, intensity profile, slice position, and preprocessing traces that can function as biometric identifiers. Near-duplicate leakage is related but broader: visually similar derived images may cross splits even when filenames or metadata no longer expose their shared origin.

Source, site/scanner, and demographic shortcuts are not leakage in the narrow train-test contamination sense, but they can produce equally misleading clinical estimates. If label prevalence differs by dataset source, scanner protocol, age, sex, or disease severity, a model can learn cohort structure instead of disease-relevant neuroanatomy. Finally, clinical-label shortcut occurs when a benchmark mostly separates advanced atrophy from healthy controls; such performance may be real for that narrow severity contrast while still failing to support early diagnosis or deployment claims.

Figure 1 visualises the four leakage modes that share the train–test-contamination structure and that motivate the edge design of the SplitGuard-AD leakage graph in §2.2. The empirical

decomposition in §3.5.1 shows that the dominant mode on the AD MRI cohorts studied here is longitudinal same-subject leakage (panel b), and that subject-only splitting alone (Protocol B) captures ∼ 93% of the correction on the ADNI headline cohort. On Tier 1 that protocol is not available: grouping by the only identifier the release carries still leaves 70% of the test images with a participant seen in training (§2.3.1). The additional component-safe layer (Protocol C) contributes a smaller marginal above that baseline, because on cohorts with intact subject identifiers the two protocols produce the same partition. What separates them is what happens when those identifiers are not intact: §3.8 measures the divergence directly, and finds that Protocol C prevents 49% of the residual patient leakage Protocol B admits once a tenth of the identifiers are gone.

2.2 The audit instrument

SplitGuard-AD is an open-source reference implementation and audit pipeline, not a general-purpose library: it comprises five sequential modules (Figure 2).

2.2.1 Metadata registry

Each image is assigned a row in a manifest with fields including `image_id`, `subject_id`, `subject_id_confidence`, `raw_class_label`, `binary_label`, `source_dataset`, and preprocessing lineage. For public JPEG benchmarks lacking original DICOM metadata, subject IDs are inferred from filename conventions with confidence flags for unparseable cases; Tier 1 shows that such inference can be wrong in ways no confidence flag detects (§2.3.1).

2.2.2 Leakage-graph construction

A leakage graph $G = (V, E)$ is constructed over the image set V. We distinguish the edge families that are *blocking* (those whose endpoints are merged into one component and therefore can never be split) from those that are *reported* for inspection, because the released implementation treats them differently and the distinction changes what the framework guarantees.

Blocking edges. An edge (u, v) is added, and the endpoints merged, if u and v share an inferred subject identifier, a session identifier, a source volume (identified by series UID), an acquisition date within a subject, or are byte-identical (equal SHA-256). Which of these are available is cohort-dependent, and the released audit reports the per-reason edge counts so a reader can see which rules actually fired: the ADNI tier supplies subject, session, series-UID, longitudinal and acquisition-date identifiers; the public-benchmark tier has no acquisition provenance at all, so only inferred subject and exact-SHA edges exist there. Supplementary Table S5 gives the per-cohort breakdown.

The practical consequence is stated plainly: on the two source cohorts the connected components coincide with participant grouping, and on Tier 1 they coincide with an identifier that is not participant identity. The additional edge families are implemented, auditable and do fire, but none of them changes a single component here. Establishing their value requires a cohort whose leakage structure departs from subject identity (§4.4).

Reported edges. Near-duplicate pairs are detected with a difference hash (dHash) at 16×16, i.e. a 256-bit code, and flagged when the Hamming distance satisfies $d \leq \delta$ with $\delta = 4$. Under the default policy (`--near-duplicate-policy review`) these pairs are written to a candidate file and surfaced in the audit report but are *not* merged; this is the policy

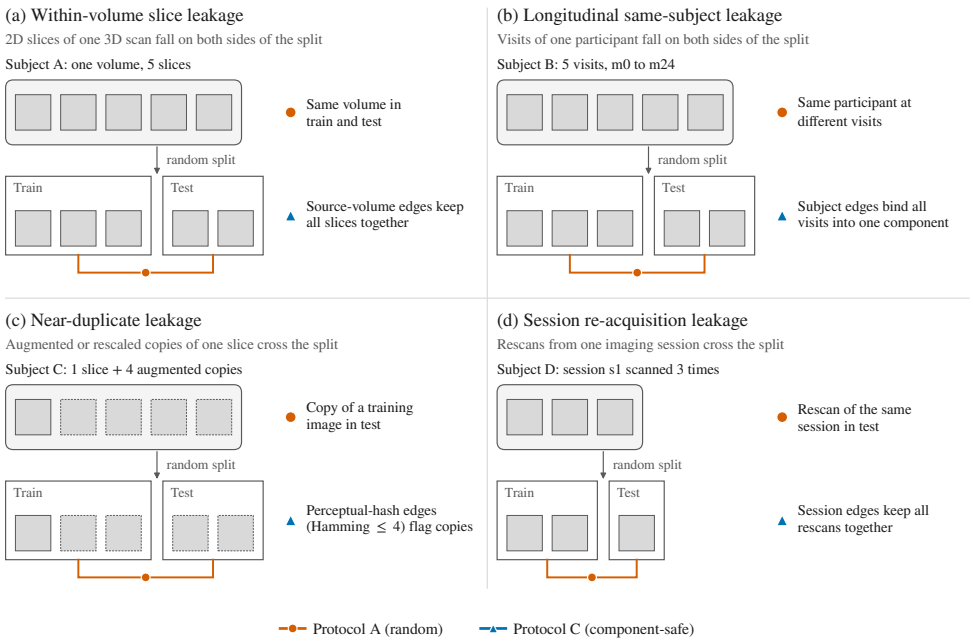


Fig. 1: Four leakage modes in slice-based AD MRI benchmarks, and the SplitGuard-AD edge type that addresses each: (a) source-volume (series-UID) edges, (b) subject-identifier edges, (c) near-duplicate dHash (Hamming ≤ 4, reported not merged under the default policy), (d) session-identifier edges. In each panel the bracket and circle mark the leak a random split (Protocol A) produces, and the triangle the edge that prevents it under component-safe splitting (Protocol C). Connected components under the union of the blocking edges are the indivisible split units. **This figure describes the framework's design space, not its empirical exercise in this paper.** On the two source cohorts every connected component contains exactly one participant; on Tier 1 the components built from the release's own identifier contain up to three participants each, because that identifier is not a participant identifier (§2.3.1). Modes (a) and (d) do fire on ADNI (209 source-volume and 223 session pairs, contributing to 115 of the 382 components), but each joined a pair that mode (b) had already joined, so no component differs from a plain subject grouping; mode (c) is detected and reported but not merged under the default policy. All four are implemented and auditable; none is load-bearing on these cohorts, which is why the component layer's measured marginal over subject-wise splitting is small (§3.5.1, Supplementary Table S5). Demonstrating them requires a cohort with re-acquisition or augmentation structure that subject identity does not already capture; we name this as the primary validation gap in §4.4.

under which every manifest and every number in this paper was produced. A `blocking` policy that additionally merges them is implemented and switchable. We keep review as the default because the identities recovered for Tier 1 (§2.3.1) let this threshold be calibrated rather than assumed, and the calibration shows that no setting of it is safe to gate on. At $d \leq 4$ the detector is precise against recovered identity (precision 0.986) but sees only 1.6% of same-participant

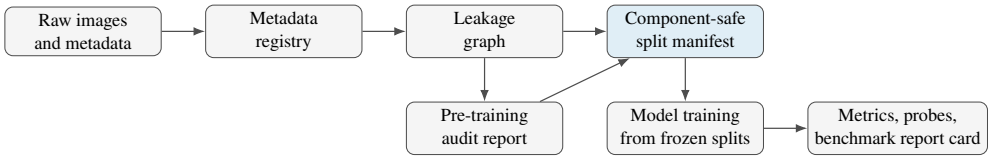


Fig. 2: The SplitGuard-AD pipeline: from raw images to audit-verified split manifests and benchmark report cards. Modules are executed sequentially; each produces a versioned artefact consumed by the next.

pairs, so merging its firings moves the Tier-1 test contamination from 70.0% to 61.2%: real, and nowhere near a fix. At $d \leq 8$, where it recovers 82% of adjacent-slice pairs, its 8,636 firings merge the 6,400 images into 27 components, so the split stops separating anything at all. A further 1,343 of those firings join images carrying opposite labels and are refused rather than merged, because the splitter requires label-pure components: the rule is simultaneously too aggressive to split on and unable to act on a sixth of what it finds. A rule with that shape belongs in an audit report a human acts on, not in an automatic gate.

We report the hash family and width explicitly because they are not interchangeable: dHash at 256 bits with $\delta = 4$ is a far stricter criterion than the 64-bit perceptual hashes for which the $\delta = 4$ convention was originally proposed. Connected components $C_1, \ldots, C_k$ of G define the atomic units for splitting: no component may be split across train, validation, and test partitions.

2.2.3 Component-safe split generation

Given target split ratios $r_{\text{train}} : r_{\text{val}} : r_{\text{test}}$ and a class stratification target, SplitGuard-AD assigns entire components to splits by stratified assignment against image-count targets, seeded for reproducibility: within each stratum it fills the validation target with whole components, then the test target from what remains, and leaves the rest to training. Targets are image counts rather than component counts, and the test target is the remainder after train and validation are rounded, so the three never over-subscribe the stratum. On Tier 1 the subset choice is an exact dynamic-programming solve; on ADNI it is a greedy descending-size fill approaching the target from below. Ties are broken on a frozen seeded rank, which is what makes the manifests both byte-reproducible and genuinely seed-sensitive. The resulting split manifest is frozen as a CSV file and versioned; no training script may discover or modify splits at runtime. Algorithm 1 formalises the full procedure: leakage graph construction (lines 1–8) followed by component-safe assignment (lines 9–18). All randomness is funnelled through the single SEED argument, so the manifest is byte-reproducible.

Implementation note

Algorithm 1 shows the double loop in $O(N^2)$ form for clarity; exact identifier edges are implemented by hash-grouping in $O(N)$. The near-duplicate comparison is the exception, evaluated over all $\binom{N}{2}$ pairs, which is tractable at the scale studied here ($N \leq 6{,}400$) and is the component that would need locality-sensitive indexing at 10^5 images or more.

Algorithm 1 SplitGuard-AD leakage-graph construction and component-safe split generation.

Require: image set V; subject-ID map σ; session-ID map ς; source-volume (series-UID) map ν, each defined only where the cohort supplies it; dHash distance $d(\cdot,\cdot)$ (256-bit code) with threshold $\delta = 4$; near-duplicate policy $\rho \in \{\textsc{review}, \textsc{blocking}\}$ (default review; see §2.2.2); target ratios $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}})$; class label $y : V \to \mathcal{Y}$; RNG seed s

Ensure: split assignment $\pi : V \to \{\text{train}, \text{val}, \text{test}\}$ such that no leakage component is split across partitions

$E \leftarrow \emptyset$; $R \leftarrow \emptyset$ ▷ E: blocking edges; R: reported-only pairs
for each pair $(u, v) \in V \times V$ with $u \neq v$ **do**
  if $\sigma(u) = \sigma(v)$ **or** $\varsigma(u) = \varsigma(v)$ **or** $\nu(u) = \nu(v)$ **then**
    $E \leftarrow E \cup \{(u, v)\}$ ▷ subject, session, or source-volume edge, always blocking
  else if $d(u, v) \leq \delta$ **then**
    if $\rho =$ blocking **then**
      $E \leftarrow E \cup \{(u, v)\}$
    else
      $R \leftarrow R \cup \{(u, v)\}$ ▷ surfaced in the audit report, not merged
    end if
  end if
end for
$\{C_1, \ldots, C_k\} \leftarrow$ connected components of $G = (V, E)$
// Assign whole components, stratified, by image-count targets
rng $\leftarrow$ RandomGenerator(s)
for each stratum S (class label; on ADNI also field-strength bucket) **do**
  Freeze a seeded shuffle rank on the components of S
  $m \leftarrow \sum_{C \in S} |C|$ ▷ targets are image counts, not component counts
  $m_{\text{tr}} \leftarrow \text{round}(r_{\text{tr}} m)$; $m_{\text{val}} \leftarrow \text{round}(r_{\text{val}} m)$; $m_{\text{te}} \leftarrow m - m_{\text{tr}} - m_{\text{val}}$
  $B_{\text{val}} \leftarrow$ ChooseSubset(S, m_{val})
  $B_{\text{te}} \leftarrow$ ChooseSubset($S \setminus B_{\text{val}}, m_{\text{te}}$)
  $B_{\text{tr}} \leftarrow S \setminus (B_{\text{val}} \cup B_{\text{te}})$ ▷ remainder; never chosen explicitly
end for
// ChooseSubset(S, m): on ADNI, visit S by $|C|$ descending with ties broken on the frozen rank and take each component that still fits in the remaining budget, approaching m from below; on Tier 1, solve the same objective exactly by dynamic programming over attainable sums.
$\pi \leftarrow$ assignment induced by $(B_{\text{tr}}, B_{\text{val}}, B_{\text{te}})$
return π, audit report {component sizes, class balance, cross-partition contamination, source/site/demographic associations}

Protocol nomenclature

Throughout the experiments we evaluate three split protocols on the same image universe:

Protocol A (Random, leaky). Random image-level shuffle into train/val/test; the conventional benchmark setup.

Protocol B (Subject-only). Split by the supplied subject identity: all images of one subject land in one partition, with no component or near-duplicate constraint. Reported where it isolates a distinct layer, chiefly the ADNI experiments of §3.3.
Protocol C (Component-safe, SplitGuard-AD). Full leakage-graph component-safe split: subject identity, scanner-session re-acquisitions, and near-duplicate components are all respected.

The three protocols form a strict ordering (A ⊇ B ⊇ C) in terms of leakage permissiveness; the three-point degradation analysis in §3.5.1 traces the contribution of each layer.

2.2.4 Pre-training audit

Before training, SplitGuard-AD emits a structured audit report with three parts: (i) component overlap across partitions; (ii) label distribution per partition; and (iii) association between label and each audited covariate, given as Cramér's V per partition. The covariates are site code (extracted from the ADNI `PTID`), scanner field strength and sex.

This last part is complementary to scanner-effect work [36] and to harmonisation in the ComBat lineage [37–39]. SplitGuard-AD surfaces such confounds but does not remove them, because the gap measured here is dominated by the subject-identity channel, which post-hoc harmonisation does not address. The clinical claim-boundary statement in the report card (§4) is authored, not emitted by the audit. The audit report issues a binary GO / NO-GO decision for training according to the formal criteria in Supplementary Table S6.

2.2.5 Model audit layer

After training, SplitGuard-AD evaluates trained model representations with a linear probe: probe accuracy above chance indicates residual shortcut encoding. In this release only the *subject-identity* probe is implemented and reported (§3.5, `scripts/run_biometric_probe_adni.py`), because subject identity is the channel the component-safe split is designed to close. Probes for dataset source, site/scanner, and demographic attributes follow the same construction and are the natural extension of this layer, but are not implemented in v1 and no such probe result is reported here; the site–label association we do report (§3.3) is a metadata-level contingency statistic, not a representation-level probe.

2.3 Cohorts

Cohort roles

Two source cohorts carry the measurements. The **ADNI1: Complete 3Yr 1.5T** collection (§2.3.3) is the paper's *headline* cohort: all headline numerical claims in the abstract, the cost-of-leakage analysis, the subgroup analysis, and the dose-response stress test derive from it. **OASIS-1** (§2.3.2) is the replication cohort, where participant identifiers are authoritative. It is operationally external to ADNI: a different recruiting site, scanners, acquisition protocols and recruitment era. The direction-preservation we report between the two cohorts is therefore a partial cross-site generalisation check, not external validation in the regulatory sense. The third tier is not a third cohort. The **Kaggle public JPEG benchmark** (§2.3.1) is, as we show by content matching, a redistribution of OASIS-1 itself whose participant identifiers have been replaced by a filename index that both merges and splits participants. We therefore use it for

what it documents rather than for what it would replicate: an observed and measurable instance of the provenance damage that §3.8 otherwise has to simulate. Throughout the experiments we use the shorthand *Tier 1* (redistributed public benchmark), *Tier 2* (OASIS-1), *Tier 3* (ADNI1 headline) for brevity.

2.3.1 Tier 1: a redistributed public benchmark

The widely distributed public Kaggle Alzheimer's MRI benchmark dataset [40] comprises 6,400 JPEG images of uniform 176×208 pixels in four severity classes: NonDemented (n = 3,200), VeryMildDemented (n = 2,240), MildDemented (n = 896), and ModerateDemented (n = 64). The release states no origin, carries no acquisition metadata, and exposes subject identity only through a parenthesised filename index (*e.g.*, `NonDemented_patient100_1 (100).jpg`).

What the images are

We recovered the origin by content. Every image was matched against axial slices of all 436 OASIS-1 cross-sectional volumes in their processed, atlas-registered, masked form (`T88_111/*_masked_gfc`; `scripts/audit_tier1_ground_truth.py`). All 6,400 images match unambiguously: the lowest correlation between an image and its matched slice is 0.9996, while the highest correlation any image reaches with a slice of a *different* participant is 0.9623. The benchmark is 200 OASIS-1 participants at 32 consecutive axial slices each (atlas slices 71–102). The folder labels reproduce those participants' Clinical Dementia Rating exactly, for every one of the 6,400 images (CDR = 0 → NonDemented, 0.5 → VeryMild, 1 → Mild, 2 → Moderate), and no participant's slices are split across two folders: they are CDR, not the weak folder-derived labels they are usually assumed to be. Every one of the 200 participants is also in our Tier-2 OASIS-1 set, so Tier 1 and Tier 2 are two views of one cohort, and the cross-tier replication in this paper rests on two source cohorts rather than three.

What redistribution destroyed

The filename index is not a participant identifier. Its 324 distinct keys divide into 149 that correspond to exactly one participant and 175 that merge two or three participants under a single key, while 179 of the 200 participants are spread across two or more keys and one is spread across 32. A further 128 images (2.0%) carry no parsable index at all. Grouping by this key is therefore not subject-wise splitting: it is grouping on a key that is wrong in both directions at once, and the supplied identifier alone does not reveal that. Recovering it required matching the images against the source cohort, which a user of the release is not placed to do.

Against the recovered identities the damage is exact. In the frozen seed-42 split used throughout this paper, which groups by the filename key, 85 participants have images in more than one partition, 46 of the 58 test participants also appear in training, and 647 of the 960 test images (67.4%) belong to a participant the model was trained on. Table 2 repeats that measurement for every grouping rule the release supports.

Near-duplicate structure, read against ground truth

No exact SHA-256 duplicates exist. The near-duplicate detector (dHash at 16×16, a 256-bit code, Hamming $d \leq 4$) flags 1,582 candidate pairs, of which 1,559 (98.5%) join two slices

Table 2: Every grouping rule the Tier-1 release supports, measured against the recovered participant identities over five seeds. Contamination is the share of test images whose participant also appears in training. The filename key, the only identifier the release provides, leaves seven test images in ten contaminated. Near-duplicate edges at $d \leq 4$ remove an eighth of that; at $d \leq 8$ the graph collapses (8,636 pairs merge the corpus into 27 components, the largest holding 3,117 images, while a further 1,343 firings join opposite-label images and are refused rather than merged), which is a different failure rather than a fix. Only the recovered participant identity yields a clean split.

Grouping rule	Components	Largest (images)	Test contamination mean [min, max]	Test participants in train / total
Filename key (release identifier)	324	32	70.0% [62.0, 78.6]	46.4/59.4
+ near-duplicate $d \leq 4$	219	192	61.2% [53.4, 67.5]	42.0/56.2
+ near-duplicate $d \leq 8$	27	3,117	100%	1.0/1.0
Near-duplicate $d \leq 8$ alone	999	1,770	99.5% [99.0, 100]	104.4/105.0
Recovered participant	200	32	0%	0/29.0

of the same participant and 1,526 join slices one position apart in the same volume. Of the 57 pairs that straddle the frozen partitions, the recovered identities make 52 the same participant and 5 different. The filename key had called 36 of those 57 "different subjects", and in 31 of them the key was wrong and the detector right. Read against the release's own identifier the detector looks like a 63% false-positive rate; read against ground truth its cross-partition output is 91% true leakage. An audit that can only compare a detector against the identifiers it already holds will systematically mistake a broken identifier for a false alarm.

Calibrating the detector

With every pair's identity known the threshold is calibrated over all 20,476,800 pairs rather than argued for. At $d \leq 4$ it is precise and sparse (precision 0.986, recall 0.246 of adjacent-slice pairs, 0.016 of all same-participant pairs, false-positive rate 1.1×10^{-6}); at $d \leq 8$ it is sensitive and unusable as a blocking rule (precision 0.699, recall 0.816, false-positive rate 1.3×10^{-4}, the 8,636 pairs that collapse the graph in Table 2). The median Hamming distance between two slices of one participant is 33 bits against 35 between participants, so the detector separates neighbouring slices, not participants, and perceptual hashing cannot stand in for a participant identifier.

2.3.2 Tier 2: OASIS-1

We use OASIS-1 as the metadata-rich replication cohort [41]; LaMontagne *et al.* [35] describe the successor OASIS-3 longitudinal release for context. All 12 official raw cross-sectional archives were available locally, together with the demographic/clinical and reliability spreadsheets. The clinical spreadsheet contains 436 session rows from 416 unique subjects, including 20 MR2 reliability sessions. For each session, SplitGuard-AD selectively extracts one processed masked T88 MPRAGE Analyze pair (`PROCESSED/MPRAGE/T88_111/*_masked_gfc.img/.hdr`) without fully expanding the raw archives.

Table 3: ADNI1: Complete 3Yr 1.5T cohort demographics. Subject-level summary; subjects are assigned to a diagnosis group by highest-priority label observed across all visits (AD > CN > MCI). Age computed at each subject's earliest scan from `PTDOBYY`; baseline MMSE is each subject's earliest non-missing `MMSCORE`.

	CN	MCI	AD	Total
Subjects (n)	142	68	172	382
Scans (n)	794	442	946	2182
Age, years (mean ± SD)	75.7 ± 5.4	75.1 ± 7.1	75.0 ± 7.3	n/a
Female (n, %)	68 (48%)	15 (22%)	76 (44%)	n/a
Baseline MMSE (mean ± SD)	29.2 ± 1.0	27.5 ± 1.7	24.7 ± 2.5	n/a

CDR is used as the primary clinical label: CDR = 0 is mapped to NonDemented, and CDR > 0 to Demented. Sessions with missing CDR are retained in the manifest for auditability but excluded from the primary classification experiment. The resulting OASIS-1 manifest contains 135 NonDemented sessions, 100 Demented sessions, and 201 sessions with missing CDR. Five central coronal slices are derived from each processed volume, yielding 2,180 total slices and 1,175 labelled slices (675 NonDemented, 500 Demented). The frozen OASIS-1 SplitGuard-AD split contains 825/175/175 labelled slices in train/validation/test, with 165/35/35 subjects respectively and zero subject or component overlap across splits.

2.3.3 Tier 3: ADNI1 Complete 3Yr 1.5T (headline cohort)

We use the ADNI1: Complete 3Yr 1.5T image collection as the paper's headline cohort, accessed under an active LONI IDA Data Use Agreement; the consortium overview [42], the MRI-acquisition protocol [43], and the clinical-characterisation core [44] provide the foundational methodology references for the ADNI release used here. All headline numerical claims in the abstract, the intro, and the cost-of-leakage and dose-response analyses derive from this cohort. The collection contains 2,182 pre-processed 1.5T T1-weighted volumes (MPR or MPR-R variants with GradWarp, B1 correction, N3, and intensity scaling already applied) from 382 unique subjects spanning the ADNI1/GO/2/3/4 phases (Table 3). A small subset (28 scans, 17 of which fall in the CN/AD universe) are magnetization-transfer T1 (MT1) acquisitions rather than MPRAGE; these are retained in the primary analysis and excluded in a pre-specified sensitivity arm (§3.3).

Diagnosis labels are taken from the ADNI diagnostic summary table (`DXSUM_PDXCONV_ADNIALL`) under a phase-aware visit-level mapping. Every visit in the export we use carries the canonical `DIAGNOSIS` code (1=CN, 2=MCI, 3=AD), and all 2,180 resolved labels in the released manifest come from it (`label_source=DIAGNOSIS`). The released builder also implements two fallbacks for exports that differ from ours: a longitudinal `DXCHANGE` collapse to the destination state, and the mutually exclusive ADNI1 legacy flags `DXAD`, `DXMCI`, `DXNORM`, which some ADNI1 case report forms use instead of one categorical field. Neither fallback fires on this export, which has no `DXCHANGE` column at all, so neither contributes a label reported here; we describe them because they are in the released code, not because they shaped these results. Scan-to-visit alignment uses a 180-day

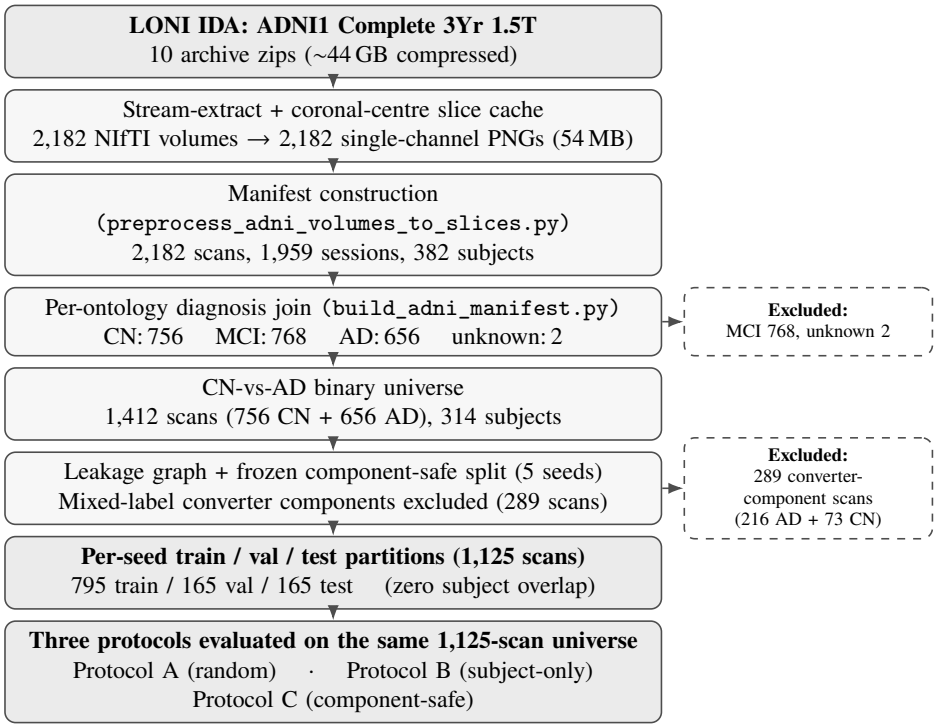


Fig. 3: ADNI1: Complete 3Yr 1.5T data flow from LONI IDA archive download to per-seed train/validation/test partitions. Each transition annotates the row count and the script responsible (`preprocess_adni_volumes_to_slices.py`, `build_adni_manifest.py`, `make_adni_splitguard_split.py`). Dashed boxes on the right record exclusions: MCI / unknown diagnoses excluded from the CN-vs-AD task by construction, and 289 CN/AD scans (216 AD + 73 CN) from mixed-label longitudinal converter components excluded from the primary arm and re-included in the sensitivity arm (§3.3).

date-proximity join through the MRI metadata table, with a documented date-only fallback when the viscode-keyed lookup fails. The full mapping rule and edge-case handling are pre-specified before any model training; the resulting manifest yields 756 CN, 768 MCI, and 656 AD scans, plus two scans whose diagnosis cannot be resolved (kept in the manifest with `diagnosis_group=unknown` for auditability and excluded from the primary CN-vs-AD task).

Linkage-integrity audit

Every ADNI label rests on this join, so it is accounted for per scan in `adni_linkage_audit.csv` (`scripts/build_linkage_audit.py`; schema and provenance in Supplementary Material S5). Of the 2,182 scans, 1,695 (77.7%) were labelled through a direct `(RID, VISCODE)` match, 485 (22.2%) through the 180-day date-proximity fallback, and 2 (0.1%) remained unresolved. The primary arm is the worst case: of its 1,123 CN-vs-AD scans, 841 carry the exact visit key and 282 (25.1%) the date fallback, against

297 of 1,412 (21.0%, with 1,115 on the exact key) in the whole CN-vs-AD universe, because the converter components this arm excludes are almost fully covered by the visit key (5.2% date-only). That one label in four rests on a date match is a limitation of the label chain, so the exact-linkage arm of §3.3 retrains all three protocols on the 841, keeping the same 220 participants, and asks whether the headline gap survives without the weaker quarter.

For apples-to-apples comparison with Tier 2, each volume is reduced to a single 2D coronal-centre slice. The slice is computed once at ingest with 1st–99th percentile intensity normalisation and saved as a single-channel PNG; this caching step is byte-identical to the volume-time computation in §2.3.2 but fits the full collection in 54 MB rather than 77 GB of NIfTI volumes. The leakage graph links scans sharing a subject identifier, a session identifier, a series UID, or an acquisition date; these four rules fire on 1,800, 223, 209 and 222 pairs respectively. The ADNI builder computes no image-hash edges, by design and not by omission, so this tier carries no near-duplicate protection: that channel is measured on Tier 1, where the images are public and may be hashed and redistributed. The 382-subject collection collapses into 382 single-subject components (mean 5.7 scans per component, range 4–11), so the frozen SplitGuard-AD split is effectively a subject-stratified partition with additional session-, series- and date-level edges that, on this cohort, every one of which is subsumed by the subject rule (§3.8).

The CN-vs-AD binary task universe excludes 768 MCI scans by construction (the binary task is cross-sectional, not longitudinal progression-prediction) and a further 289 scans (216 AD + 73 CN) drawn from components tagged `mixed_label`: subjects whose visit sequence crosses CN, MCI, and AD diagnoses (longitudinal converters). Including those subjects in the primary arm would require either (i) breaking the subject-level partition (the same subject contributes CN visits to one split and AD visits to another), or (ii) collapsing their longitudinal trajectory to a single subject-level label, which conflates cross-sectional CN-vs-AD classification with progression-prediction. We adopt the cleaner option: exclude these 289 scans from the primary arm and re-include them in a pre-specified sensitivity arm (§3.3, "Sensitivity to converter exclusion"), assigning each mixed-label component a CN-or-AD majority label for stratification only. No scan is relabelled: training and evaluation use each scan's own visit-level diagnosis throughout, so a converting participant contributes CN scans as CN and AD scans as AD within one component. The total inflation gap in the sensitivity arm (+0.142) is *larger* than in the primary arm (+0.118), so the exclusion does not drive the headline result. Removing these 289 scans leaves 1,123 CN/AD scans. The frozen split manifests carry 1,125 rows, because two scans whose visit-level diagnosis did not resolve (`diagnosis_group=unknown`, §2.3.3) are kept in the manifest and flagged there rather than deleted from it. They are not, however, evaluated: every trainer maps only CN and AD to a target and silently skips any other value, so the two rows enter no partition the model sees. The evaluated universe is therefore 1,123 scans across 220 single-subject components, equivalently 220 participants, since every component here contains exactly one subject. Under component-safe splitting each frozen seed produces train/validation/test partitions of 793/165/165 evaluated scans; the manifest row counts are two higher. We record the discrepancy rather than reconciling it silently, because a manifest row count and an evaluated scan count are different quantities and conflating them is how a cohort size drifts. Figure 3 summarises the end-to-end pipeline from the 10 LONI IDA archive zips through the slice-cached manifest to the per-seed splits.

3 Results

Statistical conventions

Unless stated otherwise, intervals are $B = 10{,}000$ paired-seed bootstrap intervals [45, 46] over a five-seed array. At $n = 5$ they describe seed-level variability rather than population-level uncertainty, and the "95%" label is kept only for tabulation convention. Where we say a direction holds, we mean the sign of the inter-protocol delta is the same in every one of the five seeds; we do not quote the resample count, because resampling five same-signed numbers reproduces that sign with probability one and so adds nothing to the seed count itself. A two-level hierarchical (subject × seed) bootstrap is run on every arm from its per-image predictions and reported alongside; it is the more defensible inferential quantity and is roughly 1.9–4.7× wider across the arms. We weight, in decreasing order, effect size, sign consistency, the hierarchical interval, and last the paired-seed width.

Training recipe and compute

The three 2D cohorts and both 2D backbones share one recipe: ImageNet-pretrained ResNet-18 (or DenseNet-121 in the architecture arms), $3\times224\times224$ input with ImageNet mean/std normalisation, horizontal flip and $\pm7^\circ$ rotation augmentation, AdamW with cosine annealing, weight decay 10^{-2}, class-weighted BCE loss and one-shot test-set evaluation. Cohort-specific settings are learning rate 3×10^{-4} (Tier 1, OASIS-1) or 10^{-4} (ADNI); 30, 20 and 15 epochs; batch size 32 on Tier 1 and OASIS-1 and 16 on ADNI; seeds $\{42, 0, 1, 2, 3\}$ on Tier 1 and OASIS-1 and $\{0, 1, 2, 3, 4\}$ on ADNI. The volumetric arm departs from that recipe in three ways that matter for reading it: a 3D ResNet-18 on 128^3 volumes at batch size 4 for 15 epochs, trained from random initialisation rather than ImageNet weights, on z-scored volumes with no augmentation. Tier 1 additionally trains under CUDA mixed precision, which the other arms do not use. An early version of the volumetric trainer computed $1-\text{AUROC}$ from a descending rank sum, which would have selected the least-trained epoch of every run; it was corrected before this arm was trained, and the released code carries a regression test for it. Every run was executed in one code state with deterministic cuDNN kernels and seeded data loaders, on one NVIDIA H100 except the dose-response matrix, rerun on an NVIDIA RTX PRO 4500, the volumetric arm, run on an NVIDIA RTX 2000 Ada, and the exact-linkage, structured-corruption and label-permutation arms, run on an NVIDIA RTX 4090 (§Declarations); the full configuration is in Supplementary Table S1.

3.1 Observed provenance damage on a redistributed benchmark

Setup

Tier 1 is trained under three protocols that differ only in what they group by. Protocol A shuffles images at random. Protocol B′ groups by the filename key, the only identifier the release carries and therefore the strongest protocol an audit run from inside the release can implement. Protocol C′ groups by the participant identity recovered in §2.3.1. Everything else is held fixed. The backbone is ResNet-18 [47] with ImageNet pre-trained weights, trained for 30 epochs at batch size 32 with random horizontal flip and $\pm7^\circ$ rotation. The optimiser is AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, AMSGrad disabled) at learning rate 3×10^{-4} and weight decay 10^{-2}, with cosine annealing and a class-frequency-weighted BCE loss. The seed array is $\{42, 0, 1, 2, 3\}$.

Table 4: Tier-1 test AUROC under the three grouping rules, five seeds. Protocol B′ groups by the filename key the release ships, which is the best an audit inside the release can do; Protocol C′ groups by the participant identity recovered from OASIS-1. The gap column is A−C′.

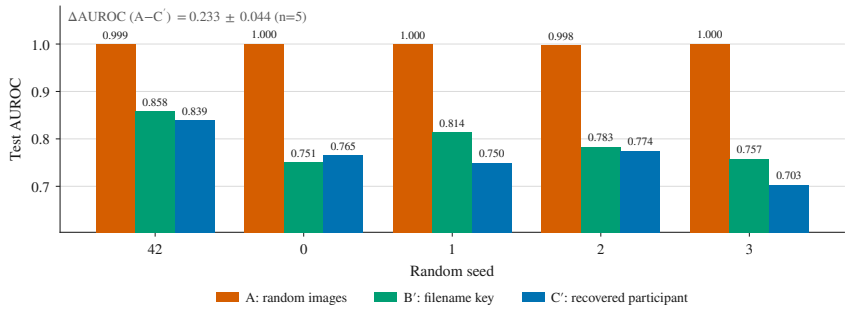
Seed	A: random	B′: filename key	C′: participant	Gap (A−C′)
42	0.9994	0.8581	0.8387	+0.161
0	0.9999	0.7508	0.7655	+0.234
1	0.9996	0.8142	0.7496	+0.250
2	0.9976	0.7834	0.7743	+0.223
3	0.9999	0.7574	0.7030	+0.297
Mean ± SD	0.999 ± 0.001	0.793 ± 0.044	0.766 ± 0.049	+0.233

Leakage verification

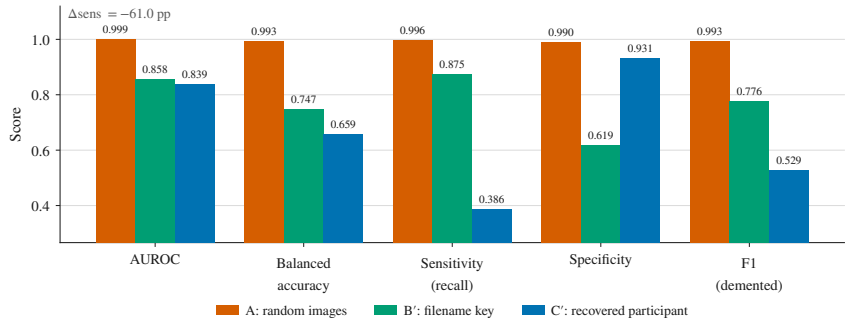
Under Protocol A, 195 of 219 test-set filename keys (89.0%) also appear in training. Under Protocol B′ no filename key crosses a partition, so an audit restricted to the release's own metadata reports zero leakage, while 70.0% of the test images in fact belong to a participant the model trained on. Under Protocol C′ the contamination is zero (Table 2).

Results

Random splitting inflates test AUROC from 0.766 to 0.999, a gap of +0.233. That part is the familiar result. The part this tier exists for is the middle column: grouping by the release's own identifier recovers 88.6% of that gap and leaves +0.027 AUROC of inflation standing, because the key it groups on merges and splits participants. A practitioner who does everything correctly with the information the benchmark provides still reports an inflated number, and nothing inside the benchmark reveals it. This is the difference between leakage as a protocol error, which the literature already documents, and leakage as a provenance failure, which survives a correctly applied protocol. Figure 4 separates what is stable across seeds from what is not. Protocol A sits above both grouped protocols in all five seeds (Figure 4a) and on every metric at seed = 42 (Figure 4b), so the headline gap is not a seed artefact. The residual between B′ and C′ is the smaller quantity and it is not uniform: their AUROC ordering reverses in one seed of five (Table 4), and at seed = 42 C′ is the more specific of the two while B′ is the more sensitive, which is what a threshold applied to a less separable score produces. The +0.027 residual is therefore a five-seed mean and not a per-run guarantee, and we hold it to the same standard as the ADNI component marginal rather than a softer one: its subject × seed interval is $[-0.058, +0.113]$, which spans zero, exactly as ADNI's 0.008 [-0.033, +0.042] does. Both are unresolved at five seeds and we report both that way. What the Tier-1 tier establishes is therefore the *contamination*, which is structural and needs no interval – 70% of test images share a participant with training under the release's own key – not the AUROC consequence of that contamination, which these five seeds cannot pin down. The distinction matters because the contamination is what a practitioner can check and the AUROC consequence is what they would have to take on faith.



(a) Per-seed test AUROC under the three grouping rules.



(b) All test-set metrics at seed = 42.

Fig. 4: Tier-1 under random splitting (A, vermilion), the release's filename key (B′, green) and recovered participant identity (C′, blue): (a) seed stability, and (b) the per-metric breakdown. Panels are stacked vertically rather than side-by-side so that the in-figure tick labels and bar annotations remain legible at print size.

What the three test sets are not

A random split spreads a participant's 32 slices over all three partitions, so its test partition holds many more participants, each contributing far fewer images, than the participant-grouped one. That asymmetry is a property of the protocols and cannot be removed by resampling: subsampling participants leaves AUROC essentially unchanged, because it is a rank statistic over pairs. What it does mean is that the three AUROCs are estimated over differently composed test sets, and we report the composition rather than adjusting for it (§3.3, Supplementary Table S8).

Learning curves

Figure 5 shows validation AUROC per epoch. Protocol A saturates within the first third of training, which is the signature of memorising images it will be tested on rather than learning a disease feature; Protocol B′ tracks it at a lower level, still lifted by the participants its key failed to group; Protocol C′ converges lowest and slowest.

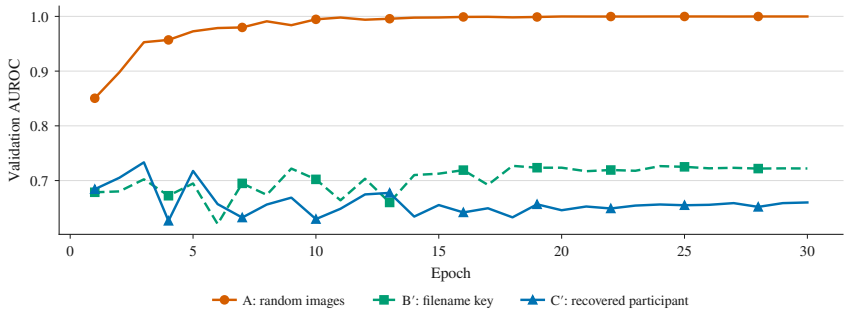


Fig. 5: Validation AUROC per epoch at seed = 42 for Protocol A (random, vermilion), Protocol B′ (filename key, green) and Protocol C′ (recovered participant, blue). The vertical distance between the first and last curves is the inflation the release's own identifier cannot remove.

Tier-1 takeaway

Tier 1 is not an independent cohort and we no longer treat it as one: it is 200 OASIS-1 participants redistributed without their identifiers. That makes it the one place in this paper where provenance damage is observed rather than simulated, and it supplies the quantity the simulation in §3.8 is calibrated against: an identifier that is wrong in both directions leaves +0.027 AUROC of inflation in place after a correctly applied subject-wise split. The mechanism behind that inflation is examined in §3.5, the layer-by-layer decomposition in §3.5.1, and the cohorts with intact identifiers in §3.2 (OASIS-1) and §3.3 (ADNI).

3.2 Replication on OASIS-1

We repeat the leaky-versus-SplitGuard-AD comparison on OASIS-1 using the same ResNet-18 family and the derived 2D slice representation described in §2.3.2. Protocol A randomly splits the 1,175 labelled slices by image; Protocol C uses the frozen subject-component-safe SplitGuard-AD split. Across five random seeds, Protocol A places nearly every test subject in training (mean test-subject contamination 99.54% ± 0.68%, range 98.5–100%).

Why the OASIS-1 gap is mechanically smaller than ADNI's

Before we report the AUROC numbers, we surface the structural properties of the OASIS-1 slice manifest that bound the leakage surface *a priori* (script: `scripts/oasis_subject_level_summary.py`; JSON: `reports/tables/oasis1_subject_level_summary.json`). Across the labelled subset ($n = 235$ subjects with binary NonDemented/Demented labels; a further 181 subjects are dropped for missing CDR at the labelling stage) OASIS-1 has 5 coronal slices per session and 2,180 slices across 436 sessions. Sessions per subject average 1.05: only 20 subjects, 4.8% of the 416 imaged, carry a repeat session. Per-subject image counts are in fact comparable between the two cohorts (OASIS-1 $\approx$ 5.2 slices per subject; ADNI1 primary arm $\approx$ 5.1 scans per subject at one coronal-centre slice per scan, §2.3.3), so the smaller OASIS-1 gap is not attributable to a thinner per-subject image stack. The more plausible mechanism is structural: only 4.8% of OASIS-1 subjects carry a repeat session, so the cohort contains almost no longitudinal within-subject variation for a model to exploit, whereas the ADNI1 cohort is longitudinal by

Table 5: OASIS-1 replication: random slice split against the participant-safe SplitGuard-AD split, mean ± SD over five ResNet-18 seeds at 20 epochs.

Metric	Protocol A	Protocol C	Gap (A − C)
AUROC	0.828 ± 0.029	0.793 ± 0.020	0.035 ± 0.037
Balanced accuracy	0.712 ± 0.027	0.704 ± 0.024	0.008 ± 0.043
F1 (demented)	0.650 ± 0.069	0.683 ± 0.015	−0.033 ± 0.076
Sensitivity	0.687 ± 0.225	0.771 ± 0.137	−0.084 ± 0.220
Specificity	0.738 ± 0.174	0.638 ± 0.179	0.100 ± 0.185
Test subjects leaking	99.54%	0%	n/a

construction. The OASIS-1 replication is therefore best read as a directional replication of the mechanism, not as a high-powered replication of the inflation magnitude reported on ADNI (§3.3). The OASIS-1 gap is reported at the image level, and should be read as the weaker of the two cross-cohort quantities when quoted alongside the ADNI subject-level result.

The OASIS-1 result (Table 5) validates the core leakage mechanism on the cohort Tier 1 was drawn from: the leaky protocol still places nearly every test subject in the training set (99.54%). While the operating-point metrics (F1, sensitivity) are statistically noisy across five seeds due to the smaller dataset size, and one seed produces a negative AUROC gap, the contamination result confirms that random slice splitting creates severe subject leakage even when true metadata are available.

DenseNet-121 architecture replication

Under a DenseNet-121 backbone and the same five seeds the OASIS-1 gap is +0.048, mirroring the ResNet-18 result above in point estimate and in seed-to-seed noise on this small cross-sectional cohort.

3.3 Replication on ADNI1, and its robustness arms

We repeat the leaky-versus-SplitGuard-AD comparison on the ADNI cohort introduced in §2.3.3, using the same ResNet-18 family and coronal-centre-slice representation as §3.2. Three split protocols are compared on the same 220-participant CN-vs-AD manifest of 1,125 rows, of which 1,123 are diagnosis-resolved. Protocol A splits images at random 70/15/15 and is the leaky protocol. Protocol B partitions by subject 70/15/15, without the leakage-graph component, series or near-duplicate constraints. Protocol C uses the frozen SplitGuard-AD component-safe split. Test partitions hold 169, 169 and 165 scans respectively, because the component-safe subset-sum assignment cannot hit the target size exactly. Each protocol is run for five seeds, 15 epochs each, with AdamW at lr = 10^{-4} and a class-balanced BCE-with-logits objective.

Under Protocol A, the random image-level split places nearly every test subject into the training partition: the mean test-subject contamination across the five seeds is 99.7%, with several seeds reaching 100%. Protocols B and C, by construction, achieve 0% test-subject overlap.

Table 6: ADNI inflation gap, 5 seeds, 15 epochs, ResNet-18, coronal centre slice. Numbers are mean AUROC with paired-seed bootstrap 95% CI (B=10,000). "Test contam." = mean % of test subjects also appearing in the train partition. The lower panel drops the 28 magnetization-transfer T1 scans for a sequence-homogeneity sensitivity check.

Protocol	AUROC [95% CI]	n_{test}	Test contam.
All scans (MT1 included):			
Random (leaky)	0.947 [0.926, 0.965]	169	99.7%
Subject-only	0.837 [0.810, 0.862]	169	0.0%
Component-safe (SplitGuard)	0.829 [0.813, 0.846]	165	0.0%
Total gap (random − component-safe)	0.118 [0.096, 0.145]		
Subject-leakage (random − subject-only)	0.110 [0.086, 0.139]		
Component-marginal (subject-only − comp.-safe)	0.008 [-0.033, 0.042]		
MT1 acquisitions excluded:			
Random (leaky)	0.960 [0.947, 0.973]	168	99.7%
Subject-only	0.833 [0.805, 0.857]	168	0.0%
Component-safe (SplitGuard)	0.831 [0.810, 0.855]	163	0.0%
Total gap (random − component-safe)	0.128 [0.097, 0.159]		
Subject-leakage (random − subject-only)	0.127 [0.103, 0.154]		
Component-marginal (subject-only − comp.-safe)	0.002 [-0.017, 0.028]		

Table 6 reports the per-protocol test AUROC with paired-seed bootstrap intervals (B=10,000 resamples of the five-seed array, descriptive; see "Statistical conventions" at the start of §3). The total inflation gap (Protocol A − Protocol C) is +0.118 [+0.096, +0.145] AUROC, and the leaky->-SplitGuard-AD direction holds in all five seeds. A two-level hierarchical (subject × seed) re-analysis, the more defensible inferential quantity, gives +0.118 with an interval of [+0.061, +0.181] (see Robustness analyses below). The gap decomposes into a large subject-identity component (Protocol A − Protocol B, 0.110) and a small marginal beyond patient-level splitting (Protocol B − Protocol C, 0.008 [-0.033, +0.042]) whose interval touches zero.

Subject-level vs image-level evaluation (recommended clinical reading)

An image-level AUROC weights participants by scan count, and participants do not contribute equal numbers of scans. We therefore aggregate each participant's per-scan predictions (mean y_{prob}) and recompute AUROC at the participant level on the converter-inclusive arm (Table 7): the gap survives the aggregation, and per-protocol AUROC is marginally higher under it, consistent with within-participant noise reduction. The image-level number is retained as the cross-cohort-comparable headline; *but the subject-level number is the recommended reading for clinical interpretation*, because it is a stricter aggregation that rules out per-subject scan-count heterogeneity as a confound and is closer to the patient-level unit of clinical decision-making. Downstream users of SplitGuard-AD report cards should quote the subject-level gap in clinical-adjacent contexts and the image-level gap only in benchmark-comparison contexts.

Table 7: ADNI image-level against subject-level AUROC on the converter-inclusive arm (five seeds). Both columns are AUROC with a paired-seed bootstrap interval at $B = 10{,}000$. Subject-level numbers average each participant's per-image predictions before computing AUROC.

Protocol	**Image-level**	**Subject-level**
Random (leaky)	0.957 [0.946, 0.969]	0.966 [0.962, 0.969]
Subject-only	0.789 [0.730, 0.851]	0.818 [0.771, 0.873]
Component-safe (SplitGuard-AD)	0.815 [0.795, 0.835]	0.829 [0.794, 0.858]
Inflation gap (Protocol A − Protocol C):		
Image-level	+0.142 [+0.131, +0.158], positive in 5/5 seeds	
Subject-level	+0.137 [+0.108, +0.173], positive in 5/5 seeds	

Table 8: ADNI robustness arms; the primary arm excludes converters. Columns are the total gap (Protocol A−C), which is the paper's headline, the subject part (A−B) and the component marginal (B−C), which is its null result; all three are measured on the same runs. Intervals are paired-seed bootstrap at $B = 10{,}000$.

Arm	**A−C**	**A−B**	**B−C**
Primary	+0.118 [+0.096, +0.145]	+0.110 [+0.086, +0.139]	+0.008 [−0.033, +0.042]
Converter-inclusive	+0.142 [+0.131, +0.158]	+0.168 [+0.105, +0.232]	−0.026 [−0.100, +0.048]
MT1 excluded	+0.128 [+0.097, +0.159]	+0.127 [+0.103, +0.154]	+0.002 [−0.017, +0.028]
DenseNet-121	+0.146 [+0.134, +0.161]	+0.136 [+0.107, +0.165]	+0.011 [−0.026, +0.053]
Size-balanced	+0.136 [+0.098, +0.159]	+0.110 [+0.086, +0.139]	+0.026 [−0.040, +0.068]
Exact label linkage	+0.117 [+0.072, +0.166]	+0.093 [+0.039, +0.136]	+0.024 [−0.017, +0.065]

Robustness analyses

Four pre-specified arms test whether the headline gap depends on a particular methodological choice, a fifth row reports the post-hoc size-balanced control of §4.4, and a sixth isolates the label-linkage question raised in §2.3.3 (Table 8). All four preserve the leaky->-SplitGuard-AD ordering in every paired-seed bootstrap resample, and the total gap moves by at most 0.028 AUROC across them. The component marginal does not behave that way. It is Protocol B − Protocol C throughout, and it inverts to -0.026 on the converter-inclusive arm, opposite to the direction a leakage-removal effect would take, since there the component-safe constraints raised AUROC relative to subject-only splitting rather than lowering it. No interval excludes zero in any arm. We therefore report the total gap as robust and the component marginal as unresolved at five seeds, and we do not resolve it by adding seeds. Two results say why. §3.8 shows the margin is near-zero on intact cohorts by construction, so the informative axis is provenance loss rather than seed count; and §3.4 measures what a spurious contrast of this kind looks like at this seed count, finding one of +0.094 between two protocols that had nothing to learn.

The sixth row answers the label-linkage question directly. Restricting the cohort to the 841 scans whose diagnosis comes from an exact visit key, which discards a quarter of the primary

arm's scans while keeping all 220 participants, gives a total gap of +0.117 against the primary arm's +0.118, positive on all five seeds, and a component marginal of +0.024 whose interval again spans zero. The headline gap is therefore not an artefact of the date-matched labels. We put that no more strongly than the arm supports: dropping a quarter of the scans widens the interval by nearly half again, from [+0.096, +0.145] to [+0.072, +0.166], so the two point estimates agreeing to 0.001 is a coincidence inside overlapping intervals rather than a precise replication. What the arm establishes is consistency, and that the date-matched quarter is not carrying the result. The arm also completes a sensitivity analysis that §4.4 could only run halfway. That one rescored the primary arm's test partition on its exact-key scans, which tests the labels the gap is measured *against* while the date-matched scans stay in training; here they are absent from both sides, and the answer does not change.

The DenseNet-121 arm additionally replicates on OASIS-1 (+0.048, §3.2), so the inflation is consistent in direction across two convolutional backbones on both source cohorts, though on OASIS-1 neither backbone's hierarchical interval excludes zero. A two-level subject × seed hierarchical resampling widens the total-gap interval on every arm, and we give all of them rather than only the primary one: [+0.061, +0.181] (primary), [+0.081, +0.208] (converter-inclusive), [+0.072, +0.191] (MT1 excluded), [+0.087, +0.209] (DenseNet-121), [+0.070, +0.209] (size-balanced), [+0.145, +0.335] (volumetric) and [+0.046, +0.195] (exact label linkage). All seven exclude zero. The two OASIS-1 arms are the exception and are reported with the cross-cohort comparison in §3.2: [−0.033, +0.107] on ResNet-18 and [−0.021, +0.121] on DenseNet-121, both spanning zero. Calibration is far worse under both honest protocols than under the leaky one, and not ordered between them (Brier/ECE: Protocol A 0.089/0.079, Protocol B 0.247/0.253, Protocol C 0.196/0.188) [48, 49]: the leaky protocol is calibrated to its own contaminated test distribution, which is not a property worth having [50]. Test-set demographic composition is balanced by construction across the five frozen splits (mean 50% ± 6% female, 49% ± 7% aged ≥ 76; 220 participants, median age 76).

Whole-volume input

The volumetric arm asks whether the shortcut belongs to the 2D slice or to the participant. A 3D ResNet-18 on 128^3 volumes, trained on the same 220 participants under the same three protocols, puts the total gap at +0.235 [+0.174, +0.298], positive in 5/5 seeds and [+0.145, +0.335] under the subject × seed hierarchical resampling, against +0.118 on the 2D primary arm. We report the two side by side rather than as a ratio: the recipes differ in pretraining and augmentation as well as in dimensionality, so the comparison establishes that the shortcut survives whole-volume input, not that volumes leak twice as much. The gap widens from both ends, and only one of those ends is about leakage. The leaky level rises to 0.971, against 0.947 in 2D, which is what a memorisable unit of one whole volume rather than one slice predicts. The honest level falls to 0.736, against 0.829 in 2D, and that is not a leakage effect: this network is trained from scratch on 703–723 volumes per run, where the 2D arms fine-tune ImageNet features, so its honest performance is not a competitive volumetric baseline and we do not offer it as one. The transferable quantity is the gap and not the level, here as everywhere else in this paper. The decomposition reproduces the 2D pattern: the subject layer carries the inflation (+0.268 [+0.163, +0.372]) while the component layer adds nothing resolvable at five seeds (−0.033 [−0.185, +0.122], spanning zero, with subject-only at 0.703 falling below component-safe just as it does in the converter-inclusive 2D arm).

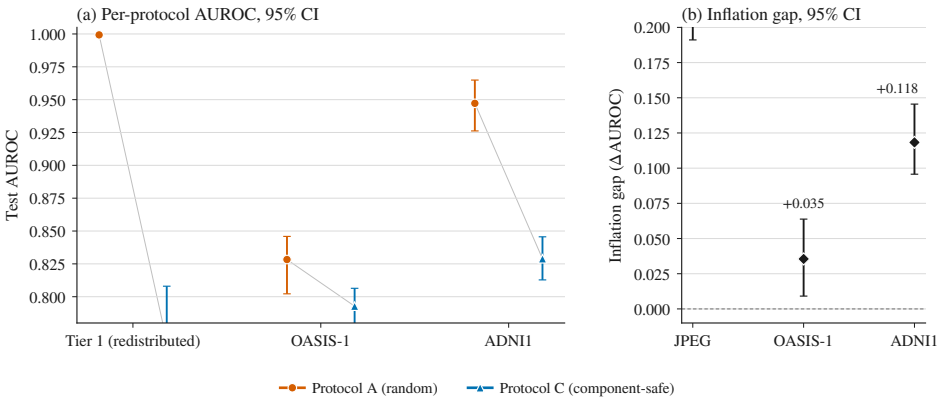


Fig. 6: Direction-preserved inflation gap on the two source cohorts and on the redistributed benchmark (ResNet-18). (a) Per-protocol test AUROC (random versus participant-safe) for each cohort; error bars are paired-seed bootstrap intervals ($B = 10{,}000$, $n = 5$ seeds per cohort; descriptive of seed-level variability, see "Statistical conventions" in §3). Grey lines connect the two protocols within each cohort. (b) The inflation gap ΔAUROC with the same descriptive intervals. On Tier 1 the participant-safe protocol is the identity recovered in §2.3.1, not the release's own identifier, which is why its gap is larger here than the one a reader of that benchmark could measure. DenseNet-121 replications are reported in §3.2 (OASIS-1) and §3.3 (ADNI); Tier 1 was run on ResNet-18 only; the figure shows ResNet-18 for visual parity with the rest of the paper.

Site × label confound check

The ADNI CN/AD universe spans 48 recruiting sites (site code extracted from the first triplet of the ADNI `PTID`); on the full universe, uncorrected Cramér's V between site and CN/AD label is 0.36 ($\chi^2 = 178$, $p < 0.001$), reflecting the well-known fact that ADNI sites differ systematically in their CN/AD subject mix. Within the five frozen component-safe SplitGuard-AD splits, this association is preserved rather than eliminated (script: `scripts/site_scanner_confound_audit.py`; JSON: `reports/tables/adni/adni_site_scanner_confound_audit.json`). Supplementary Table S7 reports bias-corrected Cramér's V (Bergsma's small-sample correction [51]) for site×label and sex×label across the five primary-arm component-safe splits. The site–label association is moderate in train ($V \approx 0.36$), substantial in validation ($V \approx 0.48$), and strong in test ($V \approx 0.54$, ranging to 0.72 across seeds); the sex–label association is negligible in train and test but reaches $V \approx 0.34$ in the small validation partition. Scanner field strength is constant across this cohort and so cannot confound the label. The framework does not perform site-aware partitioning by design and is therefore not a substitute for site harmonisation (ComBat [37–39, 52]) or for explicit site-stratified splitting; the inflation gap we report is the subject-identity channel that harmonisation does not fix post-hoc (§2.2). A site-stratified extension of the component-safe construction is named as future work in §4.4.

Figure 6 places the ADNI gap in context under one reporting convention, paired-seed bootstrap intervals at $B = 10{,}000$. The gap is positive in the mean on both source cohorts: the small cross-sectional one (OASIS-1, +0.035 [+0.009, +0.064]) and the longitudinal one with

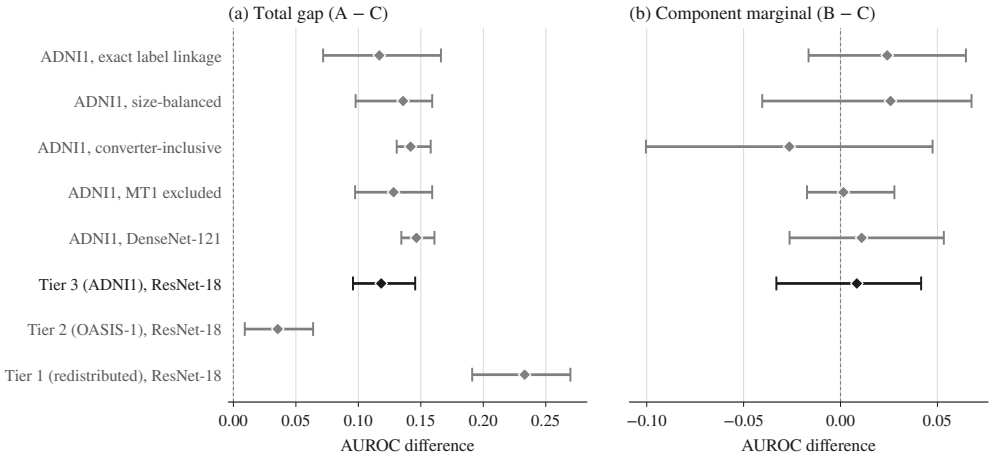


Fig. 7: The two cross-cohort arms and the six ADNI robustness arms, as point estimate and paired-seed bootstrap interval (B = 10,000, n = 5 seeds; descriptive of seed-level variability per the convention in §3). The primary arm (Tier 3, ResNet-18) is drawn in black. The volumetric arm is reported in §3.3 and not drawn here, because it changes architecture, input and training recipe at once and a shared axis would invite the comparison the text declines. **(a)** Total gap, ΔAUROC = Protocol A − Protocol C. The per-protocol AUROCs behind each arm are given in the corresponding subsection; the DenseNet-121 arms reproduce the gap rather than the absolute level (+0.048 on Tier 2 and +0.146 on Tier 3). **(b)** Component-layer marginal, Protocol B − Protocol C, on its own axis because it is an order of magnitude smaller. No interval excludes zero and the converter-inclusive arm inverts; the cross-cohort arms report no marginal and leave the slot empty. The two panels are the paper's headline and its null result on the same runs. The ADNI hierarchical (subject × seed) bootstrap, the more defensible inferential cross-check, gives gap +0.118, CI [+0.061, +0.181], direction preserved.

substantial repeat structure (ADNI, +0.118 [+0.096, +0.145]). It is positive again, at +0.233 [+0.191, +0.270] (hierarchical [+0.159, +0.311]), on the redistribution of the first, where each participant contributes 32 slices rather than five. The three differ in how firmly, and we state the difference rather than averaging over it. On ADNI and Tier 1 the sign holds in every seed and both the paired-seed and the hierarchical intervals exclude zero. On OASIS-1 it does not: one of five seeds is negative (−0.010), and while the paired-seed interval excludes zero the subject × seed hierarchical interval is $[-0.033, +0.107]$, with the sign preserved in 84% of resamples. OASIS-1 is therefore a directional replication at the level of the seed mean, not an independently significant one, and we read it that way throughout. Patient identity is the dominant leakage channel on the two cohorts where a subject-only protocol exists to decompose it, ADNI and Tier 1; OASIS-1 carries no such arm. Subject to that, the random-split benchmark overstates AD-classification performance by a magnitude robust to the seed array, the converter-inclusion rule, the acquisition type, the CNN backbone, input dimensionality and the strictness of the label linkage.

Figure 7 consolidates the headline inflation results across both source cohorts, the redistributed benchmark, both backbones and every ADNI robustness arm in a single reporting frame.

3.4 Patient reuse manufactures apparent discrimination

The gap of the previous section is real and robust, but on its own it cannot separate "the leaky protocol learned the disease better" from "the leaky protocol recognised its patients". Both produce a higher AUROC under random splitting, and no amount of replication across arms distinguishes them. We separate them by removing the disease.

Design

The instrument is not new. Chaibub Neto *et al.* [28] introduced permutation tests for exactly this purpose, to separate a disease signature from a participant signature when records of one participant can reach both sides of a split, and label-shuffling controls of the same shape are now standard in neurophysiology benchmarks. We apply it here because the question it answers is the one the inflation gap leaves open, and because the protocols it compares are retrained rather than re-scored.

Diagnosis labels are permuted across participants and the three protocols are retrained at the frozen settings, five seeds, 15 epochs. The permutation is applied once over the whole cohort rather than inside each partition, so a participant who straddles a boundary carries the same permuted label on both sides: within-participant label consistency survives, the association between anatomy and diagnosis does not. That detail decides what the control can answer. Permuting inside each partition preserves per-partition class balance exactly and hands a straddling participant two different labels, which forecloses identity memorisation by construction and drives every protocol to chance for a reason that has nothing to do with the split.

Whether the permutation actually took effect is checked rather than assumed. Across the scored test rows the permuted label coincides with the participant's true diagnosis 48.3%, 55.1% and 50.3% of the time under the three protocols. That is the chance rate, and it is the precondition for reading anything below as patient recall: had the true labels survived anywhere in the pipeline, the random arm would still score high and the result would read as a stronger version of the same finding rather than as a broken one.

Result

With no disease signal left to learn, the random split still reaches an AUROC of 0.878 $[0.837, 0.919]$. The two honest protocols sit at chance: 0.521 $[0.427, 0.614]$ under subject-only splitting and 0.427 $[0.320, 0.533]$ under component-safe splitting, both intervals covering 0.5. The seed-paired contrast is +0.451 [+0.357, +0.546], positive on all five seeds.

The training curves say where it comes from. All three protocols drive the permuted training loss to a comparably low value, 0.068, 0.037 and 0.045 at the final epoch, so all three memorise the training set about equally well. Only the random split's memorisation transfers: its final validation AUROC is 0.863 against 0.453 and 0.481 for the honest protocols, on a label that carries no information about any image. This is the shortcut being taken, measured rather than inferred, and it is the evidence the probe could not supply.

What the number is not

The permuted gap of +0.451 is not the inflation under real labels, which is +0.118. It measures the capacity of the memorisation channel when nothing else is learnable. Under real labels the honest protocols score around 0.83 because there is a real signal to learn, and the leaky

protocol's score is bounded above; the channel measured here is one contributor to that score rather than the whole of it. We report it as evidence that the channel exists and is large, not as a revised effect size.

A measured noise floor

The control also prices the paper's null result. Under permutation the subject-only and component-safe arms differ by +0.094 [+0.013, +0.175], an interval excluding zero, between two protocols that both have nothing to learn and are therefore both at chance. It cannot be a difference in what was learned, and its sign has a mechanism worth naming, because it is not specific to permuted labels. The trainer keeps the epoch with the best validation AUROC. With the disease signal removed, the honest arms' validation partitions carry none either, so that maximum is selection on noise, and the smaller the partition the further it overshoots: validation optimism, the best validation AUROC minus the test AUROC, is +0.007 under subject-only splitting on a validation partition averaging 33 participants, and +0.159 under component-safe splitting on one averaging 22. The leaky arm shows −0.008 on a validation partition averaging 124 participants, because it holds the same patients as training: the signal it selects on is real memorisation, so there is nothing to overshoot.

Two things follow. A contrast of +0.094 arises here between protocols with nothing to learn, an order of magnitude above the 0.008 component marginal the main experiment reports (Table 8), which is why we decline to read that marginal as an effect and decline to settle it with more seeds of the same kind. And a component-safe design pays for its safety in validation size: 22 participants is too few to select a checkpoint on, with permuted labels or real ones, which is a cost of the protocol we had not previously quantified and which any user of it inherits. It is the scale of apparent protocol-to-protocol difference this design admits at five seeds, and it is why we decline to read the component marginal as an effect and decline to settle it by adding seeds of the same kind.

3.5 Is patient identity decodable, and does that explain the gap?

The permutation control shows the shortcut is taken. A natural next question is what makes it available, and the answer is a structural property of the representation rather than anything about the protocol. A linear probe (LinearSVC) trained on frozen penultimate-layer features (512-dim) of the 15 ADNI checkpoints of the converter-inclusive arm recovers participant identity far above chance, and to the same degree whichever protocol trained the representation: 211.7× chance under component-safe splitting. Holding out a whole visit per participant instead of a random fifth of their images, so that no repeat acquisition from the same session sits on both sides, leaves identity decodable at 204.8× chance. We run the probe on ADNI rather than on Tier 1 because ADNI supplies the participant identifiers the probe classifies; on Tier 1 the release's key is not participant identity (§2.3.1).

The interpretation is narrower than the number invites. Identity is decodable from every one of these representations, including those trained under honest protocols, so the probe cannot explain a *difference* between protocols: a quantity that is the same in both conditions cannot account for the AUROC gap between them. What it establishes is a structural property of 2D MRI. Repeated slices of one head share shape, intensity profile and slice position, so identity is available to any visual representation, and that availability is what makes test-set contamination exploitable. This also qualifies the reading that saliency evidence invites: Rumala [29]

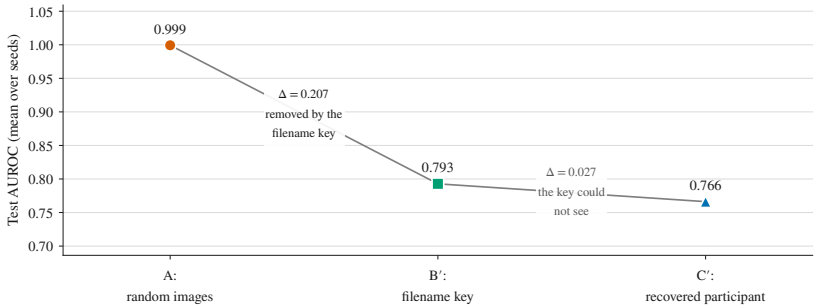


Fig. 8: Three-point degradation on Tier 1. Each step changes only what the split groups by: random images, then the filename key the release ships, then the participant identity recovered from OASIS-1. The first step is what an audit run inside the release can achieve; the second is what its lost provenance hides.

observes that a leaky network attends to subject-identifying structure and takes that as the mechanism of the inflation, but identity in the representation is a precondition for the shortcut rather than evidence that it was taken. The permutation control of §3.4 supplied that evidence by a route needing no saliency map, and the probe explains why the route exists: identity is there to be used. The probe uses one architecture and one probe family; a non-linear adversary in the manner of DeepBrainPrint [16] or BrainPrint [15] would upper-bound it. The causal weight therefore rests on the three controlled interventions of §3.4, §3.7 and §3.8.

3.5.1 Three-point degradation curve

The probe answers *whether* identity is encoded; the degradation curve answers *how much* of the inflation each grouping removes (Figure 8).

The A→B′ step is ΔAUROC = +0.206 and the B′ →C′ step a further +0.027. On ADNI, where identifiers are intact, the same decomposition puts ∼ 93% of the gap in the first step. The two cohorts bracket the claim: grouping on a correct identifier removes most of the inflation, and grouping on a damaged one does not.

Decomposition takeaway (and head-to-head against GroupKFold)

The three-point degradation isolates ∼ 93% of the ADNI inflation gap to subject-identity transfer; on Tier 1 the release's own identifier recovers 88.6% of the gap, the remainder being the participants that identifier fails to group (§3.1). Protocol B enforces the same subject-independence constraint as `GroupKFold`, under a fixed three-way split rather than K-fold cross-validation (§1); we did not run `GroupKFold` itself, so this decomposition quantifies the marginal value of the component-safe layer above subject-grouped partitioning rather than constituting a head-to-head tool benchmark. That marginal, reported throughout as Protocol B − Protocol C, is 0.008 [-0.033, +0.042] on the ADNI primary arm and -0.026 on the converter-inclusive arm (Table 8): small, of inconsistent sign, and with no interval excluding zero. On Tier 1 the comparison does not exist in that form, because the release carries no identifier that Protocol B could group on.

That marginal should be read with a further caveat that we did not appreciate until auditing our own splits. Protocols B and C do not differ only in their grouping. Protocol B shuffles

participants and cuts at 70/15/15 by participant count; Protocol C consumes a frozen manifest produced by a size-descending greedy bin-packer that hits 70/15/15 by image count. The two therefore differ in the grouping (the variable of interest) and also in the assignment algorithm and the quantity the ratios are computed over. On a difference that small, the assignment step is not obviously negligible relative to the grouping step. The same control failure, measured directly in §3.8, was worth about half the apparent effect there. We therefore report the component marginal as an upper bound on what the grouping contributes rather than as an isolated estimate of it, which reinforces rather than weakens the conclusion already stated: the marginal is small, of inconsistent sign, and not resolved.

The greedy also leaves a visible fingerprint on the frozen ADNI manifests. Because it fills the validation and test bins first and takes the largest components that still fit, the partitions end up nearly disjoint in component size: across the five seeds, training components span 4–6 scans (mean 4.65), validation 7–9 (mean 7.50) and test 5–7 (mean 6.11). Participants with longer longitudinal records are systematically routed away from training. Class balance survives this, because components are stratified by label before the subset-sum assignment runs (39.1%, 38.2% and 40.0% AD by image across the three partitions), but within each class the test partition is enriched for participants with more follow-up. That matters clinically: in this cohort AD participants contribute fewer scans than CN participants (4.55 ± 0.97 against 5.56 ± 1.00), so scan count tracks retention, and retention tracks disease course. This is an instance of the duration-dependent selection bias described in the data-handling literature [53], arriving through the splitter rather than through the cohort.

We do not leave this as a caveat. The frozen manifests are retained because every number in this paper is computed on them and exact reproducibility is worth more than a cosmetic correction, but a size-balanced split was generated specifically to test whether the headline result depends on the defect. Visiting components in shuffled rather than size-descending order removes the stratification. Mean component sizes become 5.15, 5.00 and 5.07 scans in train, validation and test, against 4.65, 7.50 and 6.11 on the frozen split. The total gap is +0.136 on the size-balanced split against +0.118 on the frozen one, with hierarchical intervals [+0.070, +0.209] and [+0.061, +0.181]. We report it beside the primary arm rather than after it. The composition defect is real, it was found by this project's own audit of its own splits, and the effect it was suspected of producing is not there. What it does mean for the audit gate, which passed this split on every check it performs, is taken up in §4.4.

The practical recommendation on a cohort like this one is therefore the familiar one, split by participant identity. The component and near-duplicate layers cost almost nothing here and buy nothing measurable either, and we claim no AUROC benefit from them on ADNI or OASIS-1. What they buy appears on the axis where it can, once identifiers begin to disappear (§3.8). Readers running the framework on a cohort with clean, converter-free identities should expect the component-safe and subject-only protocols to be indistinguishable, and should read that as the instrument reporting correctly rather than as a disappointment.

Where grouping by the available identifier is blind

Subject-wise splitting can only protect images whose participant is known, and a redistributed benchmark is exactly the case where that premise fails without saying so. Tier 1 makes the failure measurable, because its true identities are recoverable from outside the release (§2.3.1), which is not something its users have.

The exposure is not a handful of edge cases but the structure of the release, measured in §2.3.1: a key that merges and splits participants, images with no key at all, and a split that every check the release supports calls clean while most of its test set is contaminated. This is the regime the component layer exists for, and the honest reading of Tier 1 is that the layer does not rescue it either, because the redistribution removed the session and series identifiers the graph would have carried. The consequence for a reader of public benchmarks is stronger than a recommendation to split by subject: where a release does not document provenance, a subject-wise split of it cannot be verified from inside the release at all, and the number it reports cannot be interpreted.

3.6 What the gap costs at a screening operating point

Running both protocols on the same universe surfaces two findings the leaky benchmark conceals. A sex-related performance disparity emerges under both honest protocols and collapses to indistinguishable from zero under the leaky one (Supplementary Material S6); sex differences in AD are well documented [54–56], and we report ours as exploratory without attributing it to any of those mechanisms. The gap also translates into a screening-operating-point cost that a reader of the leaky number would underestimate (§3.6.1). Both rest on the per-image predictions persisted by the converter-inclusion arm.

3.6.1 Cost-of-leakage as an arithmetic illustration

To express the gap in units a clinician can weigh, we re-expressed the sensitivity difference at a fixed screening specificity of 0.90 as missed-diagnosis counts per 1,000 screened, under literature-cited prevalence anchors. Sensitivity at that specificity is 0.866 ± 0.046 under Protocol A against 0.535 ± 0.103 under Protocol C, which at the US 75–84 population prevalence of 13.8% [57] is 45.7 additional missed AD cases per 1,000 screened if the leaky benchmark is trusted.

This is deterministic arithmetic applied to the measured sensitivities, not a deployment estimate. It inherits every cohort caveat in §4.4: a single 2D coronal-centre slice, a research cohort that is not a screening population, a threshold chosen on validation, and no prospective evaluation. We give it because an AUROC difference is hard to weigh and a count is not, and we give it once. The operating-point derivation, the sensitivity of the result to the chosen specificity, and the second prevalence anchor are in Supplementary Material S2.

3.7 Leakage as a dose: injecting test-set overlap

The cross-tier inflation gap and the cost-of-leakage translation both take the leakage condition as binary (leaky vs. honest). A more informative question for benchmark designers is the dose-response: if p% of test subjects also appear in train, how much AUROC inflation should we expect? We answer this with a controlled stress-test on the converter-inclusive ADNI sensitivity arm.

Design

Starting from the Protocol C component-safe split, we generate a family of injected splits at target contamination fractions $p \in \{0, 0.25, 0.50, 0.75, 1.0\}$, where *$p$ is the share of test scans that belong to a training participant*, not the share of test participants. For each non-zero p we

draw $k = \lfloor p \cdot 258 \rceil$ participants from the multi-visit training subset and substitute one alternate scan of each into the test set, displacing an equal number of original test scans. Because each donor contributes a single scan, the share of contaminated test *participants* is never smaller and is usually far larger (63%, 77% and 86% of test participants at $p = 0.25$, 0.50 and 0.75, converging to equality only at $p = 1$), which is why every composition below is done in scan units. Class balance of the *injected partition* (118 CN / 140 AD over 258 rows) is held fixed at every overlap level by stratified substitution. The partition the model is *evaluated* on is not held fixed either, and this qualifies the calibration. The binary pipeline drops MCI scans and substitution displaces them, so the evaluated CN/AD subset grows from 203 rows at $p = 0$ to 258 at $p = 1$, with the AD share rising from 47.7% to 54.3%. These are five-seed means; the $p = 0$ count runs from 197 to 207 across seeds.

The training partition is not held fixed either, and this is the more consequential of the two. Each substituted scan is moved out of training, so evaluated training rows fall from 1,041 at $p = 0$ to 783 at $p = 1$, 24.8% of them, as the dose rises. The donor *participant* remains in training, which is why donors are drawn only from the multi-visit subset, so the identity leakage the arm is built to create is genuine; what the arm does not do is hold the training set constant while adding it.

The dose axis is therefore confounded with both test composition and training size, and the two plausibly act in opposite directions: thinning the training set should depress AUROC at high p, while the class enrichment above has no clear sign. The slope is consequently descriptive of this substitution procedure as a whole, and is not a causal price per unit of realised leakage. Both quantities are measured by `scripts/analyze_dose_response_composition.py` and reported in `reports/tables/adni/adni_dose_response_composition.json`. We report it rather than correct it because the two cannot be separated without changing the injection mechanism, and because the two cannot be separated without changing the mechanism: the slope is used only as a floor (§3.6.1). Otherwise this is biometric-identity leakage alone: no slice or near-duplicate is shared between train and test, only the patient. We run the full 5 seeds × 5 overlaps matrix on two architecture families (ResNet-18 and DenseNet-121, 15 epochs each, lr = 10^{-4} AdamW), a total of 50 training runs.

Result

Figure 9(a) shows the resulting per-seed AUROCs and the seed-averaged dose-response curves with 95% paired-bootstrap CIs. The seed-averaged curves increase with test-subject overlap for both architectures; per-seed monotonicity holds in 5/5 ResNet-18 seeds but in 0/5 DenseNet-121 seeds (Supplementary S4); the fitted ResNet-18 endpoints (0.823 at $p = 0$, 0.948 at $p = 1$) sit within 0.02 of the converter-inclusive arm's honest and leaky levels (0.815 and 0.957, §3.3), which is the arm the injected splits are built from. We read that agreement as a sanity check and not as a calibration: the matrix and those levels come from the same code state on different GPUs (§Declarations), and nothing in the injection was tuned to reproduce them. The straight-line OLS fit (Figure 9(b)) recovers

$$\widehat{\mathrm{AUROC}}_{\text{ResNet-18}} = 0.823 + 0.125 \cdot p, \qquad R^2 = 0.853, \tag{1}$$

and a similar but noisier curve for DenseNet-121 ($\widehat{\text{AUROC}} = 0.861 + 0.091 \cdot p$, $R^2 = 0.64$). The slope of +0.125 AUROC per unit overlap on ResNet-18 implies a cohort- and architecture-specific calibration point: on this ADNI universe and this 2D-slice representation, every 10 percentage points of test-subject overlap inflates the reported AUROC by approximately +0.013. The point-estimate predictions from Equation 1 are usefully read as the deterministic part of a noisy calibration; the per-seed scatter around the regression line in Figure 9(b) is approximately consistent with an additive Gaussian residual, so the dose-response can equivalently be written

$$\text{AUROC}(p) \;=\; \alpha + \beta \cdot p \;+\; \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{2}$$

with $(\widehat{\alpha}, \widehat{\beta}) = (0.823, 0.125)$ and $\widehat{\sigma} = 0.0191$ on ResNet-18 (0.0250 on DenseNet-121), the residual standard deviation of the twenty-five per-seed observations about the OLS fit. The unit matters: because the band is meant to describe a *single* training run, it has to be estimated from single runs. Taking it instead from the five per-overlap means would give 0.008 and understate the spread roughly 2.5-fold, since averaging five seeds removes most of the seed-level variance the band exists to represent. So estimated, $\widehat{\sigma}$ gives an order-of-magnitude prediction band for one run at a given p and lets the equation be used probabilistically (*e.g.*, the probability that a benchmark reporting AUROC A at measured contamination p exceeds its true component-safe value by more than 0.05 can be estimated from the upper tail of $\mathcal{N}(A - \widehat{\alpha} - \widehat{\beta}p,\ \widehat{\sigma}^2)$). We report the OLS slope as a descriptive calibration constant rather than as an inferential coefficient (the twenty-five (seed, overlap) observations are not independent under OLS). A random-intercept model fitted by iterated feasible GLS under compound symmetry gives the same slope, $\widehat{\beta} = 0.1251$, because the design is balanced; what changes is the interval. With a cluster-robust sandwich standard error carrying the finite-cluster correction and a $t(4)$ critical value, the 95% CI is $[0.0866, 0.1636]$ on ResNet-18; DenseNet-121 gives $\widehat{\beta} = 0.0906$ with 95% CI $[0.0416, 0.1396]$. Both exclude zero. Full specification, the fitted variance ratio, the seed-cluster bootstrap, and Spearman monotonicity statistics are in Supplementary Material S4. Density-choice qualifications and finer-grid trade-offs are discussed in §4.4.

Interpretation

The response is approximately linear across the whole $[0, 1]$ range, with a mild saturation near the leaky end: the ResNet-18 increments shrink across the sweep (+0.046, +0.035, +0.025, +0.020) and the DenseNet-121 mean dips at the last step (0.9403 at $p = 0.75$ to 0.9387 at $p = 1$). That is what a bounded metric does near its ceiling, not a threshold below which contamination is harmless; no threshold appears anywhere in the range.

Both architectures converge at $p = 1$ on the Protocol A level, which is the same convergence the deletion sweep produces from the other direction: total subject overlap and absent identifiers are two routes to the same condition. The DenseNet-121 curve starts higher (0.853 against 0.814 at $p = 0$) and is flatter (+0.091 against +0.125), but the mixed-effects slope intervals overlap ($[0.0866, 0.1636]$ against $[0.0416, 0.1396]$; Supplementary S4), its R^2 of 0.64 trails the ResNet-18 fit and no DenseNet seed is monotonic across all five overlaps. We run no difference test and draw no inference about backbone capacity: at five seeds the trend is carried by the seed average rather than visible within a seed.

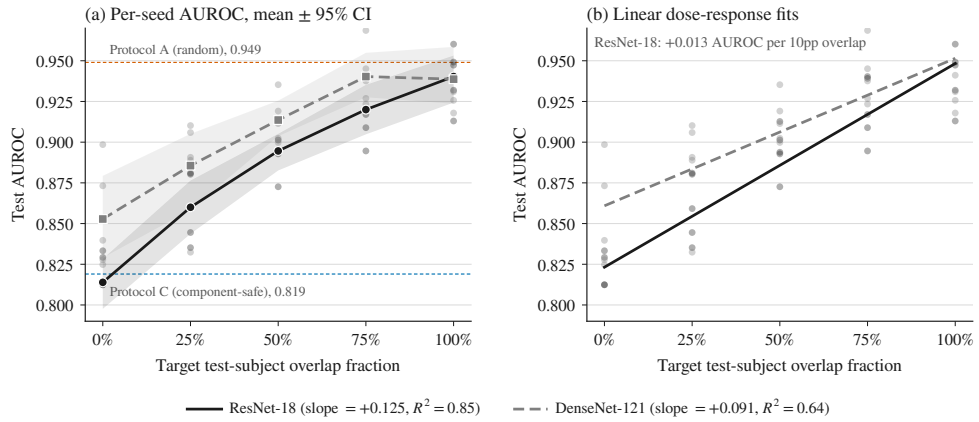


Fig. 9: ADNI leakage dose-response on the converter-inclusive sensitivity arm. (a) Per-seed test AUROCs (small dots), seed-averaged mean (markers + line) and 95% paired-bootstrap CI band for each architecture as a function of the target test-subject overlap fraction p; ResNet-18 is drawn in black and DenseNet-121 in grey (dashed), and the reference lines take the colour of the protocol they mark. Reference lines mark the *primary-arm* Protocol C component-safe baseline (0.829) and Protocol A leaky benchmark (0.947) from §3.3, shown for cross-arm comparability. The matrix itself is built on the converter-inclusive arm, whose own levels are 0.815 and 0.957; the fitted endpoints are 0.823 and 0.948 (panel a). (b) OLS linear fits to the per-seed points show the dose-response slope; the ResNet-18 slope of +0.125 AUROC per unit overlap is a cohort- and architecture-specific calibration point on this ADNI universe and this 2D-slice representation, not a claim that the slope generalises to other cohorts or 3D representations.

Practical use, and why the slope is a floor

Equation 1 converts a measured test-subject contamination fraction (which the pre-training audit already reports) into expected AUROC inflation. It should be read as a lower bound on the gap rather than as an estimate of it, for a reason that follows from how it was fitted: the injection varies test-subject overlap by substitution, so it prices the participant-identity channel and no other, while thinning the training set as it does so. A naive split on a real cohort additionally leaks through slices of one volume, near-duplicates and re-acquisitions, none of which the slope sees.

Two of the three cohorts bear this out and the third does not; we give all three. Under random splitting contamination is essentially complete on each, including 99.5% of OASIS-1 test subjects, so the equation predicts a drop of at most 0.125. Measured gaps are +0.142 on the converter-inclusive ADNI1 arm the injected splits are built from, +0.233 on Tier 1, where adjacent slices of one volume add a channel the slope does not price, and +0.118 on the primary arm, which the floor matches to within 0.01. OASIS-1 does not fit: its contamination is as complete yet its gap is +0.035, roughly a third of the prediction. We read that as the boundary of the calibration rather than a refutation, since OASIS-1 contributes five slices from one cross-sectional visit while the slope was fitted where repeat visits dominate. Used as a point estimate the equation would overstate a reader's honest AUROC, which is the wrong

direction for a quantity meant to correct over-reporting; used as a floor it answers whether the contamination present is large enough to matter.

3.8 Provenance loss: what the leakage graph buys when identifiers fail

§2.2.2 reports that on every cohort studied here the connected components coincide exactly with a subject grouping, so the component layer earns nothing over `GroupKFold`-style splitting. That is a consequence of subject being the coarsest identifier: session, series and acquisition-date edges all nest inside it, so none of them can change a component while `subject_id` is present and correct. The framework's stated scope condition is therefore that the extra edge families matter only where subject identity is absent, corrupted, or inferred wrongly. Until now that was an assertion. This section measures it.

Design

The regime is created deliberately, using the same template as the dose-response experiment in §3.7: hold everything fixed, sweep one controlled quantity, read off the curve. Here the swept quantity is provenance loss. Starting from the ADNI1 primary arm, where subject identifiers are complete and authoritative, we delete `subject_id` on a random 0/10/25/50/100% of scans and ask what each protocol does with the remainder. Deletion is applied per scan rather than per subject because that is the observed failure mode: redistributed benchmarks lose the patient index one filename at a time, so a record is typically partly recoverable rather than wholly anonymous. Under Protocol B every scan whose identifier was removed becomes its own group and may be placed in any partition. Under Protocol C the leakage graph is rebuilt from whatever identifiers survive; `session_id` and `series_uid` are globally unique on this cohort (1,003 and 1,008 distinct values, no blanks, and no value shared between two participants) and do not take `subject_id` as an input, so they continue to bind scans that `same_subject` can no longer reach.

The outcome is scored against the untouched ground-truth identifiers, which the degradation preserves out of band. We report the number of true participants with scans on both sides of a partition boundary. This is a structural measurement: it counts the leakage each protocol actually admits without routing the question through a trained model, and it is therefore free of the seed-variance and test-set-size caveats that qualify the AUROC comparisons elsewhere in this paper.

Result

Figure 10 and Table 9 give the curve. At zero loss the two protocols admit no leakage and are indistinguishable, reproducing the equivalence reported in §2.2.2. They separate immediately thereafter: at 10% identifier loss the leakage graph prevents 49% of the participant leakage subject-wise splitting admits (9.4 versus 18.6 straddling participants), and 30–37% at higher loss rates.

Both protocols are assigned by the same class-stratified bin-packer over their respective groupings, so the difference is attributable to the grouping and not to the assignment step. That control matters more than it may appear. An earlier version of this comparison assigned Protocol B by shuffling subject groups and cutting at 70/15/15 by group count, while Protocol C bin-packed by image count; under that mismatched design the graph appeared to prevent 78% of residual leakage at 10% loss. Roughly half of the apparent benefit came from the assigner

Table 9: Provenance degradation on the ADNI1 primary arm (220 participants, five seeds, mean ± SD). Columns report participants whose scans straddle a partition boundary, and p, the fraction of test-partition participants who also appear in training, the same quantity the dose-response experiment sweeps, so the two curves compose. At zero loss the protocols are identical, which is the §2.2.2 null result restated as the origin of a curve rather than as a dead end.

Identifiers	**Straddling participants**		p **(test ∩ train)**	
deleted	Protocol B	Protocol C	Protocol B	Protocol C
0%	0.0 ± 0.0	0.0 ± 0.0	0.000	0.000
10%	18.6 ± 5.0	9.4 ± 1.9	0.361	0.207
25%	31.4 ± 3.1	19.8 ± 2.0	0.374	0.408
50%	77.0 ± 5.0	53.6 ± 6.6	0.980	0.807
100%	182.6 ± 7.2	125.4 ± 1.3	0.993	0.990

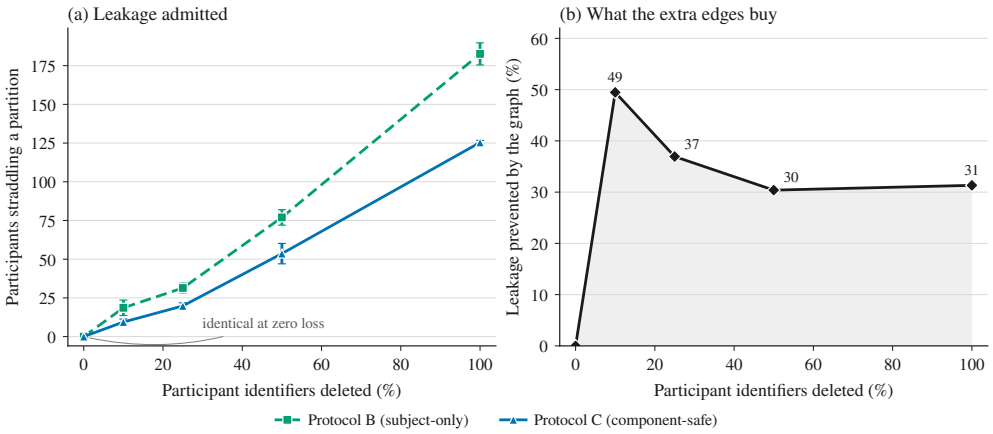


Fig. 10: Provenance degradation on the ADNI1 primary arm, five seeds, mean ± SD. **(a)** Participants whose scans straddle a partition boundary as participant identifiers are deleted. This counts admitted leakage directly, without routing the question through a trained model. The two protocols are indistinguishable at zero loss (the equivalence reported in §2.2.2, here the origin of a curve rather than a dead end) and separate immediately thereafter. **(b)** The share of that leakage the leakage graph prevents. Both protocols use the same assigner, so the difference is attributable to the grouping; under a mismatched assigner the 10% figure reads 78%. The benefit peaks in the regime where redistributed benchmarks sit and decays once too few identifiers survive for any rule to bind on.

rather than from the grouping. We report the matched comparison because only the grouping is the claim.

At the end of the sweep the protocol becomes its own failure mode

The structural counts above are what the split admits. What the experiment reports is an AUROC, and following it to the end of the sweep gives the result this paper is built on.

With identifiers complete, subject-wise splitting on this arm reports 0.771 $[0.709, 0.833]$. With every identifier removed, the same procedure, unchanged, reports 0.946 $[0.921, 0.972]$. The random image-level split of §3.3 reports 0.947 $[0.926, 0.965]$ on the same cohort. The difference between the last two is 0.001, and their intervals overlap across almost their whole length.

We state that as a failure to distinguish rather than as an equality: five seeds cannot establish that two procedures agree, and overlapping intervals are weak evidence. What the comparison does establish is that nothing in the reported number separates them. A reader handed the right-hand column cannot tell from it that the split was meant to be patient-disjoint, and neither can the procedure, which raises no error: every scan is a singleton, every singleton is its own group, and grouping by group is vacuous. The inflation across the sweep is +0.176 AUROC, and it accrues without a warning at any point.

That is the sense in which patient-wise splitting is not a property of a splitting algorithm. The algorithm is unchanged along this axis. Only the evidence it is given changes, and the guarantee it is credited with degrades continuously into the condition it was adopted to prevent.

Provenance loss amplifies non-linearly

Deleting 10% of identifiers returns 36% of test participants to the training partition under Protocol B, a roughly threefold amplification. The mechanism is elementary once stated: a participant contributes a median of five scans, and one of them losing its index is enough to detach that scan from the group and let it land opposite the other four. Participant-level protection is not degraded gracefully by scan-level provenance loss; the first missing identifier in a record breaks it.

Composing the two curves, and where the composition fails

Chaining the two curves requires them to share a unit. The injection experiment sweeps the share of contaminated test *scans*, so that is the quantity we compose through: at 10% identifier loss it is 35% under Protocol B and 18% under Protocol C, against 36% of test *participants* under Protocol B. Read through Equation 1, that is +0.043 AUROC of expected optimism under Protocol B against +0.022 under Protocol C. That composition under-predicts what the same experiment measures directly, and the comparison is worth stating because it bounds how much weight the chain carries: the provenance arm itself trains at every deletion level, and its test AUROC rises from 0.771 at no loss to 0.865 under subject-wise splitting and 0.829 under the component-safe protocol at 10% loss, that is +0.095 and +0.059 against the two composed figures just given. The chain is low by rather more than a factor of two in both protocols. Two reasons are visible: the slope is fitted on a cohort whose contamination is varied by substitution, which thins training as it adds overlap, whereas deleting identifiers perturbs both sides differently, and the slope and the shares come from different arms of one code state (§Declarations). We therefore read the composition as an order of magnitude and a lower bound, never as a coefficient, and we report the directly measured rise beside it rather than only the chain. Table 9 reports the participant share p alongside it, because that is what a split audit shows a reader.

We do not present that as a general result, because p is not monotone across the sweep. At 25% loss Protocol C's p (0.408) exceeds Protocol B's (0.374), inverting an ordering that holds at every other level and that the straddling-participant count preserves throughout. The reason is that p is a ratio whose denominator is the number of distinct test participants, and the two protocols do not produce test partitions containing the same number of them: component-safe grouping yields larger and fewer components, hence a smaller and differently composed test partition. p is therefore a defensible input to the dose-response fit, where it is measured within a single protocol, but a poor instrument for comparing protocols. The straddling-participant count, which counts events rather than ratios, does not have that defect, and it is the measurement the result above rests on. We give the composition at 10% as an illustration of what chaining the two experiments would yield, and explicitly not as a validated protocol-level comparison.

Scope

Three limits belong with this result. The measurement is primarily structural: it counts admitted leakage, and the AUROC arm trained on the same degraded manifests is reported in §4.4 rather than here. The degradation is synthetic, imposed on a cohort with complete provenance rather than observed on one that lost it, which is what makes ground-truth scoring possible and also what makes it a model of the failure rather than the failure itself. And the recovery depends on which identifiers a cohort happens to carry: ADNI1 supplies globally unique session and series keys, whereas the Tier-1 benchmark supplies neither, so the curve measured here is an upper bound on what the same protocol could achieve on a cohort stripped down to filenames.

3.9 Structured provenance corruption: where the graph earns its place

The deletion experiment removes identifiers *visibly*. A missing `subject_id` is a value a splitter can see is absent, and the honest response, treating that scan as its own group, is available to any `GroupKFold` user. Redistribution rarely fails that politely. The Tier-1 benchmark in §3.1 carries a key for every image and the key is simply wrong: it splits 179 of 200 participants across several keys and merges others under one. Nothing is missing, so nothing looks broken.

Design

We therefore apply a corruption operator $C \in \{\textsc{drop}, \textsc{split}, \textsc{merge}\}$ at intensity λ to the ADNI1 primary manifest, rebuild both groupings on the corrupted result, and score residual leakage against untouched ground truth. DROP removes a scan's identifier with probability λ; SPLIT deals a participant's scans into two pseudo-identifiers with probability λ, so one patient looks like two; MERGE relabels a random pair of participants to one pseudo-identifier, so two patients look like one. The last two mimic the observed Tier-1 failure modes and leave no field missing. These operators are controlled stress tests of identifiable provenance failure modes; they are not estimates of the prevalence or the distribution of such errors in clinical repositories, and we draw no conclusion about how often real cohorts sit at any given intensity. Only the grouping differs between the protocols compared; both use the same class-stratified assigner, for the reason given above. The experiment has two halves, and only the second needs a GPU. Contamination is a property of the split, so the leakage every cell admits is measured without

fitting anything; the AUROC each corrupted split then produces is measured by training one model per protocol per seed, 120 models in all.

Result

Table 10 gives the matrix. The three mechanisms separate cleanly, and one of them is where the component layer stops being redundant. Under DROP the graph halves the damage at $\lambda = 0.1$ (16.8 straddling participants under subject-wise grouping against 8.4), falling to 31% prevented at total loss, which reproduces §3.8 by a second route. Under SPLIT it prevents 50% at $\lambda = 0.1$ (a small absolute effect at that intensity, 1.2 straddling participants against 0.6) and 90% at $\lambda = 1$, where subject-wise grouping admits 61.6 straddling participants and the graph admits 6.4, with test-scan contamination falling from 64% to 11%. That gap is the framework's reason to exist: when one participant is dealt into two keys, `same_session`, `same_series_uid` and `same_acq_date` still join the halves, and subject-wise grouping has nothing left to join them with. Under MERGE neither protocol admits leakage at any intensity, because over-grouping two patients into one key is conservative for contamination: it destroys the participant count and the independence of the test partition without ever placing a patient on both sides. We report that null because it bounds the claim. The graph protects against identity *fragmentation*, not against every corruption, and a merged key damages a study in a way no splitter can repair.

The two leakage columns disagree in one cell, and the disagreement is instructive rather than inconsistent. At DROP $\lambda = 0.25$ the graph cuts straddling participants from 31.2 to 19.8 while the share of contaminated test scans rises from 0.364 to 0.395. The component layer builds fewer, larger groups, so a group that does straddle carries more scans across the boundary with it; counting participants and counting scans are different measurements of the same partition, and a protocol can improve one while losing the other. We print both columns rather than the one that reads better, and we compose the dose-response slope through the scan share, which is the conservative of the two (§3.7).

What the prevented contamination is worth

Preventing contamination is worth reporting only if the contamination was worth something, so we train one model per protocol per seed on each corrupted manifest and read the AUROC it produces (Table 11). The direction is consistent and the per-cell evidence is thin, and both halves of that sentence belong in the record.

Across the eight cells in which the two protocols emit different partitions, seven favour the component-safe grouping. The mean paired difference over those cells is +0.026 (SD 0.017); a one-sided sign test on the cell means gives $p = 0.035$ and a signed-rank test $W^+ = 35$, $p = 0.008$. Those two are summaries of one comparison and not independent confirmations of it, and both are descriptive rather than inferential: the cells share seeds and a base manifest, so they are not eight independent experiments. Only two cells, DROP $\lambda = 0.25$ and $\lambda = 0.5$, have an interval that excludes zero, and both exclude it by 0.001. Five seeds cannot resolve a per-cell effect of 0.02 to 0.04 AUROC when the paired SD runs from 0.024 to 0.079, so we claim a consistent direction and no per-cell effect size.

The magnitude is also smaller than the dose-response slope predicts. At SPLIT $\lambda = 1$ the graph removes 53 percentage points of test-scan contamination, from 64% to 11%, and Equation 1 would price that at roughly +0.066 AUROC against the +0.025 observed. The slope

Table 10: Residual leakage under structured provenance corruption, ADNI1 primary arm (220 participants), five seeds, cell means. Straddling counts true participants with scans on both sides of a boundary; c is the share of test scans whose true participant also appears in training, the quantity §3.7 sweeps. Both protocols use the same assigner, so a column difference is attributable to the grouping. MERGE admits no leakage at any intensity, which is why its prevented share is undefined (n/a) rather than zero: over-grouping destroys the participant count without placing a participant on both sides.

Corruption	λ	**Straddling**		**Prevented**	c **(test scans)**	
		Prot. B	Prot. C	by graph	Prot. B	Prot. C
DROP	0	0.0	0.0	n/a	0.000	0.000
	0.1	16.8	8.4	50.0%	0.337	0.177
	0.25	31.2	19.8	36.5%	0.364	0.395
	0.5	85.2	67.0	21.4%	0.991	0.891
	1	182.6	125.4	31.3%	0.989	0.979
SPLIT	0	0.0	0.0	n/a	0.000	0.000
	0.1	1.2	0.6	50.0%	0.007	0.005
	0.25	1.6	0.8	50.0%	0.018	0.004
	0.5	4.0	1.0	75.0%	0.076	0.004
	1	61.6	6.4	89.6%	0.636	0.107
MERGE	0	0.0	0.0	n/a	0.000	0.000
	0.1	0.0	0.0	n/a	0.000	0.000
	0.25	0.0	0.0	n/a	0.000	0.000
	0.5	0.0	0.0	n/a	0.000	0.000
	1	0.0	0.0	n/a	0.000	0.000

is fitted where contamination is substituted in and the training set thins as a result; corrupting provenance perturbs both sides of the split at once, and the two are not interchangeable. We report the discrepancy rather than the more favourable of the two numbers.

Two mechanisms deserve an explanation rather than a number. Under DROP $\lambda = 1$ the difference inverts to −0.004: with every participant identifier gone, both protocols admit near-total contamination, 99% of test scans under subject-wise grouping and 98% under the graph, and both saturate at 0.956 and 0.959. There is nothing left to protect, and the sign of a difference that small is noise, even though Table 10 still credits the graph with 31.3% of the straddling prevented in that cell: preventing leakage and being paid for it in AUROC are different claims, and this cell separates them. Under MERGE the difference is exactly 0.000 at every intensity because both protocols produce the same partition, so this is an identity and a determinism check, not an effect. The graph buys a modest amount of optimism back where provenance damage is the kind it can repair, and nothing at all where it is not.

Table 11: AUROC consequence of each corruption cell on the ADNI1 primary arm. Each row trains one model per protocol per seed on the *same* corrupted manifest, so the comparison is paired: the difference column and its interval are over the five per-seed differences (paired t, four degrees of freedom), not over the two marginal means. A positive difference means the component-safe grouping scored *lower*, that is, it declined optimism the subject-wise grouping accepted. The last column counts seeds in which it did. MERGE is an identity rather than a null result: when two participants share one key both protocols emit the same partition, so the same model is trained twice, and the exact zeros double as a determinism check on the pipeline.

Corruption	λ	Prot. B	Prot. C	Paired Δ	95% CI	Seeds
DROP	0.1	0.847	0.814	+0.034	[−0.062, +0.129]	4/5
	0.25	0.857	0.812	+0.045	[+0.001, +0.089]	4/5
	0.5	0.945	0.902	+0.043	[+0.001, +0.084]	5/5
	1	0.956	0.959	−0.004	[−0.034, +0.027]	3/5
SPLIT	0.1	0.850	0.836	+0.014	[−0.035, +0.064]	3/5
	0.25	0.868	0.853	+0.015	[−0.024, +0.053]	4/5
	0.5	0.845	0.806	+0.038	[−0.027, +0.104]	4/5
	1	0.877	0.852	+0.025	[−0.073, +0.124]	3/5
MERGE	0.1	0.841	0.841	+0.000	[+0.000, +0.000]	0/5
	0.25	0.806	0.806	+0.000	[+0.000, +0.000]	0/5
	0.5	0.815	0.815	+0.000	[+0.000, +0.000]	0/5
	1	0.808	0.808	+0.000	[+0.000, +0.000]	0/5

4 Discussion

4.1 Three provenance regimes

The results across this paper describe one structure, and stating it directly is more useful than restating the experiments. Patient-wise evaluation is not a property of a splitting algorithm. It is a property of the identity evidence the algorithm is given, and that evidence admits three regimes.

Regime I: intact provenance

Every observation of a participant carries a correct participant key. Here a leakage graph and a participant grouping produce the same partition, because the subject rule subsumes the others, and the component marginal is indistinguishable from zero on all six ADNI arms (Table 8). This is the design target rather than a shortfall: a provenance-aware splitter that disagreed with participant grouping where participant identity is complete and correct would be wrong. Most curated research cohorts sit here, which is why the standard advice works and why nothing in this paper argues against it.

Regime II: degraded provenance with surviving redundancy

The participant key is incomplete, fragmented or merged, but some other relation, a session, a series, an acquisition date, still joins observations of one participant. Here the two groupings separate, and the separation is what the graph is for: under identifier deletion it prevents 49% of the leakage participant grouping admits at 10% loss, and when one participant is dealt into two keys it prevents 90% (§3.9). This is the regime the paper is about, and the one the literature has not measured.

Regime III: destroyed provenance

No surviving relation connects two observations of the same participant. Tier 1 is the observed instance: the redistribution removed the session and series identifiers along with the participant one, so the component layer recovers nothing the participant key lost, and a split that respects the release's own identifier still leaves 70% of test images sharing a participant with training. This is an information limit, not a software limitation. No audit can reconstruct identity from no evidence, and we make no claim to. What an audit can do here is report that the evidence is absent, which is precisely what a benchmark's own metadata cannot do and what the three blocking checks, passing on a manifest built from a broken key, demonstrate.

The practical consequence is that the question to ask of a dataset is not whether the split was patient-wise. It is which regime the dataset is in, and that is answerable before training from the identifiers alone.

To illustrate the reporting format that SplitGuard-AD produces alongside its splits, Table 12 gives a benchmark report card for the ADNI primary-arm model under the component-safe split. It synthesises the model's claim boundary, where its identifiers come from, the blocking audit, the findings the audit reports without blocking, and the debiased performance estimates a reader would consult before treating the result as more than a benchmark score. The provenance row is the one Tier 1 motivates: the same three blocking checks pass on a Tier-1 manifest built from the release's own identifier, and only the provenance row would have shown that nothing guaranteed they meant anything.

Table 12: SplitGuard-AD benchmark report card for the ADNI1 primary-arm ResNet-18 model under the component-safe split (Protocol C), seed means over five seeds. The card illustrates the reporting format the framework produces; it is not a clinical-deployment artefact. PPV/NPV rows use the same prevalence anchors as §3.6.1.

SplitGuard-AD Benchmark Report Card	
Model Claim Boundary	CN versus AD from one 2D coronal-centre slice per scan, labels from the ADNI diagnosis table. Not intended for standalone clinical diagnosis.
Identifier provenance	ADNI `PTID`, session and series identifiers, all present for every scan.
Data Safety Audit *(blocking checks)*	
Participant Overlap	0% (✓Pass)
Session Overlap	0% (✓Pass)
Connected Component Overlap	0% (✓Pass)
Reported, non-blocking	
Label join	25.1% of this arm's labels by 180-day date proximity; the gap is unchanged on the exact-key subset (§2.3.3)
Site–label association	Cramér's $V = 0.54$ in test (Supplementary Table S7)
Component-size balance	Warning: partitions differ in scans per participant (§3.5.1)
Debiased Performance Estimates (Held-out Test, $n \approx 165$ per seed)	
Sensitivity	65.5%
Specificity	81.0%
AUROC	0.829
Balanced Accuracy	73.2%
Prevalence-conditioned predictive values (illustrative)	
PPV @ 13.8% (population) [57]	35.6%
NPV @ 13.8% (population)	93.6%
PPV @ 58.9% (memory clinic) [58]	83.2%
NPV @ 58.9% (memory clinic)	62.1%

Computed from sensitivity/specificity above; predictive values are prevalence-dependent and tracked at the same two anchors as §3.6.1.

4.2 Why contamination audits disagree about inflation

Two concurrent audits of public neuroimaging benchmarks reach apparently opposite conclusions, and the three regimes explain the disagreement without either being wrong.

Sifat *et al.* [31] examine the same Kaggle redistribution studied here and report, from the slice captions the release supplies, that its subject count does not match the labelled OASIS-1 cohort. That is the same provenance failure we characterise, reached from the identifiers the release carries rather than from the images. What a caption cannot settle is which participant an image actually belongs to, and therefore how much contamination a correct grouping still admits. Recovering that required matching the images against the source cohort, which places the finding in Regime III rather than merely flagging an inconsistency.

Vangala *et al.* [32] audit three public brain-tumour corpora and reach the stronger claim that contamination does not surface as an inflated score: removing every near-duplicate test image leaves balanced accuracy essentially unchanged. Their measurements are careful and

their patient-level figures are explicitly reported as lower bounds, because only 12.6% of their test split could be traced to a patient identifier. We read their result as compatible with ours, and the reason is a difference in what the two designs can detect.

Their comparison removes duplicated images from the test partition and re-scores a model that was trained once; no retraining is involved, so the training split still contains those images. Two things follow. A model that has memorised a participant retains that memory after the participant's duplicated images are deleted from the test set, and continues to benefit from whichever of that participant's images remain. And the identity channel is not confined to near-duplicates: in their own corpus, 28.8% of test images have a near-twin in training while 95.5% of the traceable ones share a *participant* with training. Deleting the first layer leaves the second intact. Their design therefore measures whether duplicated test items are easier to classify, which is a real and useful question, and not whether the model learned to recognise participants.

Separating those two requires an intervention on the training side. The protocol comparison of §3.3 retrains under each grouping, and the permutation control of §3.4 removes the disease signal altogether: with diagnosis permuted across participants, random splitting still reaches 0.878 AUROC while both patient-aware protocols sit at chance. A model with no pathology available to learn cannot be exploiting pathology. Vangala *et al.* offer three candidate explanations for their null and establish none of them; the permutation control discriminates between exactly these alternatives, and we report it for that reason.

The practical consequence is narrower than a disagreement and more useful. A deduplication audit, which is the field's common instrument, cannot detect participant-identity leakage, and a benchmark that passes one has not been shown to be sound. The weaker point still stands in both papers: a stable score is not evidence that an evaluation is clean.

4.3 Clinical and field-level implications

At a fixed screening specificity of 0.90, ADNI sensitivity falls from 0.866 under random splitting to 0.535 under component-safe evaluation (§3.6.1): 45.7 additional missed AD cases per 1,000 screened at the US 75–84 population prevalence anchor. A model whose sensitivity moves by a third between its benchmark and its honest evaluation is not a standalone screening tool. This gap arises from a design choice, the evaluation protocol, that is invisible in most published papers, which is the precise reason SplitGuard-AD reports the audit as a binary pre-training GO/NO-GO gate rather than as a post-hoc diagnostic. Two clinical-scope notes follow. *First*, CN-vs-AD with MCI excluded is the diagnostically easier cell of the AD imaging task. The cells that matter most for early-intervention decisions, CN-vs-MCI and stable-vs-converter MCI, are harder, and whether provenance-related inflation is larger there is an open empirical question: we did not run those tasks, and the direction is not obvious from what we did run. A smaller disease signal need not mean a larger measured gap if the leaky protocol is also less able to exploit identity. The frozen manifests released here support that experiment without rebuilding the cohort. *Second*, the headline absolute AUROC numbers (0.829 honest, 0.947 leaky) are floor estimates from a single 2D coronal-centre slice and should not be read as performance ceilings; the transferable quantity is the *gap*, not the level.

Beyond the headline number, two scoping points matter for clinical readers. First, structural MRI is not a standalone diagnostic tool for AD; in clinical practice it informs the

dementia workup by excluding structural mimics, by demonstrating atrophy patterns consistent with neurodegeneration, and by supporting clinical and biomarker-based diagnosis within a research framework that defines the disease biologically rather than by imaging alone [59]. Nor is population screening for cognitive impairment itself an established benefit [60], which is a further reason the operating-point translation in §3.6.1 is an arithmetic re-expression and not an argument for deployment, whatever the societal cost of dementia makes such a tool look worth [61]. A deep learning model evaluated on single slices labelled by a clinical rating cannot claim to diagnose Alzheimer's disease; it can only claim to classify images by that rating in a controlled benchmark setting, and SplitGuard-AD makes this distinction explicit through its clinical claim boundary and label-confidence fields. Second, lower leakage-safe performance should not be read as a failed model; it is a more honest estimate of what remains after removing patient-identity and acquisition shortcuts.

For redistributed public datasets, near-perfect internal scores are not interpretable unless the release documents provenance that makes a participant-safe split checkable: Tier 1 shows that a manifest built on a release's own identifier can pass every overlap check and still leave most of its test set contaminated. For metadata-rich cohorts such as OASIS-1, random slice splitting can still contaminate nearly every test subject, even when the resulting AUROC inflation is smaller and more seed-sensitive than on the redistributed benchmark. Future AD MRI studies should therefore release split manifests, report subject overlap checks, document where both labels and identifiers come from, include calibration and sensitivity-at-fixed-specificity reporting, and validate on cohorts whose subjects, scanners, and preprocessing pipelines are independent of training.

4.4 Limitations

- **The component layer is equivalent to subject grouping on intact cohorts, and validated beyond it only under synthetic provenance loss.** On the two source cohorts every connected component contains exactly one participant (Supplementary Table S5): the session, series-UID and acquisition-date rules fire, on 223, 209 and 222 ADNI pairs, but each joins a pair `same_subject` had already joined. The marginal over Protocol B is accordingly small and unstable: 0.008 [-0.033, +0.042] on the primary arm, and ranging from −0.026 to +0.026 across the six arms, with a sign inversion on the converter-inclusive one (Table 8). No interval excludes zero in any of them, the one measured no-signal contrast of this kind is +0.094 (§3.4), and we claim no AUROC benefit on intact cohorts. §3.8 does establish a benefit, but on a regime we imposed rather than observed: it deletes identifiers uniformly at random from a cohort with complete provenance, whereas real redistributed benchmarks may lose them with structure our model does not reproduce: by site, by archive batch, or correlated with class. That measurement is also structural rather than predictive; it counts admitted leakage; its AUROC arm, trained on the same degraded manifests, gives 0.865 under subject-wise splitting against 0.829 under component-safe splitting at 10% loss. Tier 1 supplies the observed counterpart (§3.5.1), and on it the component layer does not recover what the identifier lost, because the redistribution removed the session and series identifiers along with the participant one. A cohort whose damage leaves partial provenance for the graph to carry is still the most informative next experiment. The near-duplicate detector is now calibrated against recovered identity (precision 0.986 and adjacent-slice recall 0.246

at $d \leq 4$); it ships report-only because no threshold we measured is both safe and useful (§2.2.2).

- **Every finding is a property of a 2D coronal-centre-slice representation.** The inflation gap, the dose-response slope, the provenance curve, the subgroup disparity and the clinical translation are all measured on one slice per volume, chosen to keep ADNI and OASIS-1 comparable; OASIS-1 additionally derives five slices per processed volume, so its operating-point metrics are small-sample and seed-sensitive. ResNet-18 carries the primary results, with a pre-specified DenseNet-121 arm replicating the headline gap. Whether these quantities transfer to volumetric models is now partly answered rather than open: the volumetric arm (128^3 volumes, 3D ResNet-18, `scripts/train_adni_3d.py`) runs against the same frozen partitions and puts the gap at +0.235 (§3.3), so the identity shortcut survives whole-volume input and is in fact larger there. What does not transfer is the *level*: that network trains from scratch on 722 volumes while the 2D arms fine-tune ImageNet features, so its honest AUROC of 0.736 is a weak volumetric baseline and the arm settles the direction of the effect, not the achievable performance of 3D models. A pretrained or larger-cohort volumetric replication remains open. Multiple-instance aggregation over per-slice predictions and transformer backbones remain open, and reuse the same frozen manifests.
- **Label and identifier provenance differs by tier.** Tier-1 labels are CDR, but its participant identities are recovered by content matching rather than supplied, and we release the matching script against OASIS-1 rather than the mapping itself. Because all 200 Tier-1 participants are also in Tier 2, the two tiers are not independent replications of each other. On ADNI, 22% of labels (21% within the CN/AD subset, and 25.1% within the primary arm, which excludes the converter components the visit key covers best) are assigned through a 180-day date-proximity join rather than an exact visit key (§2.3.3), and 28 magnetization-transfer T1 acquisitions (17 in the CN/AD universe) are non-MPRAGE; a pre-specified arm dropping them yields +0.128 [+0.097, +0.159] against +0.118 [+0.096, +0.145]. Restricting evaluation to the test scans whose label comes from an exact visit key gives a gap of +0.113; that tests the labels the gap is scored against, not the ones it was trained on, since the date-proximity scans remain in the training partition. The exact-linkage arm closes that half by retraining on the 841 exact-key scans alone (Table 8), and the total gap is unchanged, so neither the labels the models learned from nor the labels they were scored against carry the result.
- **The intervals are descriptive, not inferential.** Five seeds support paired-seed bootstrap intervals that propagate between-seed variation only. The hierarchical subject × seed bootstrap, which also propagates participant-level sampling error and yields intervals 1.9–4.7× wider depending on the arm, is run on every arm. The three protocols also do not place comparable numbers of patients in the test partition (Supplementary Table S8): Protocol A averages 160.8 distinct participants per seed, Protocol B 48.0, and Protocol C 38.0, over test partitions of similar image count. Protocol C's AUROC is therefore estimated on roughly a quarter as many independent patients as Protocol A's, and part of its wider seed-to-seed spread reflects that rather than any property of the split rule. Subsampling participants does not correct for it, since AUROC is a rank statistic whose expectation barely moves with the number of clusters; the subject-level analysis above is the aggregation that does address it. The same asymmetry qualifies our recommendation of subject-level aggregation: averaging 5.35 predictions per patient under Protocol C removes substantially more variance than

averaging 1.32 under Protocol A, so it does not neutralise the difference between protocols so much as redistribute it. Any single interval here should be read descriptively rather than as inference over a hypothetical larger seed population.

- **The secondary analyses support less than they appear to.** The identity probe uses one architecture and one probe family and reports a value that does not differ between leaky and honest protocols; it establishes that identity is decodable, across sessions as well as within them, not that it causes the gap. The sex-related disparity rests on five seeds and one cohort. The clinical translation is anchored to a single fixed-specificity operating point and to US-population prevalence, is computed on ADNI only, and is an arithmetic re-expression rather than a deployment estimate. The dose-response slope is specific to this cohort, this representation and the subject-substitution injection mechanism; the DenseNet-121 fit is noisier ($R^2 = 0.64$ against 0.85) and non-monotonic at $p = 1$, so we present it as a same-direction check rather than an independent confirmation, and we do not claim the slope generalises.
- **Disjointness is necessary but not sufficient, and our own gate demonstrates the gap.** The frozen ADNI splits pass every check the audit performs, zero subject, session and component overlap, while being stratified by component size (§3.5.1). A split can be provably leakage-free and still be a poor split, and the gate as specified could not tell the difference. We regard this as the most useful negative result about the framework here, because the framework's own output produced it. The released audit now reports the component-size distribution per partition and warns when partitions are not drawn from comparable participants; against the frozen manifests it fires immediately. It warns rather than blocks, deliberately: the manifests are frozen and every result rests on them. The size-balanced regeneration reported beside the headline shows the gap survives the correction. The same audit surfaces a second composition defect we did not fix: on the converter-inclusive arm the validation partition carries 20.7% AD images against 41.3% in training and 37.5% in test, in every seed, and the operating-point and cost-of-leakage thresholds are selected on that partition. The released audit now separates the two kinds of finding: partition disjointness is the only blocking check, and component-size and class-mix imbalance are reported as advisory warnings beside the verdict rather than folded into it. Promoting them to blocking is the first item of future work, and would retroactively fail the manifests this paper reports on, which is why it has not been done here.

  The protocol also pays for its safety in validation size. Holding whole components together leaves the component-safe validation partition with about 22 participants, against 33 under subject-only splitting and 124 under a random split, which makes checkpoint selection noisy: with the disease signal permuted away, the best validation AUROC exceeds the test AUROC by +0.159 under component-safe splitting against +0.007 under subject-only. That is the no-signal worst case. On the runs this paper reports, where validation carries real signal, selection optimism is +0.026, +0.030 and −0.026 under the three protocols, every interval spans zero, and the spread is 0.055 – small, not ordered in the direction the objection requires, and negative for the component-safe arm. A fixed-epoch replication would settle it more directly and we did not run one. A practitioner adopting component-safe splitting on a cohort this size should nonetheless expect checkpoint selection to be noisy.

The gate certifies partition disjointness and nothing more. A pass does not imply site- or scanner-independent generalisation: the site–label association in Supplementary S7 survives the component-safe split, which is the central caveat of this work. Patient identity is one shortcut channel among several, and closing it leaves site, scanner, protocol, acquisition date and demographic structure open.

5 Conclusions

Patient-wise splitting is not a property of a splitting algorithm. It is a property of the participant identifier the algorithm is given, and we measured that dependence across its full range on one cohort in one code state. With identifiers complete the procedure reports 0.771 AUROC; with every identifier removed the same unchanged procedure reports 0.946, against 0.947 for the random image-level splitting it exists to prevent. The intervals overlap across almost their whole length, the inflation across the sweep is +0.176, and no step of the procedure raises an error at any point along it. A study that reports having split by patient has reported a procedure, not a guarantee, and the distance between the two is what this paper measures.

The endpoints themselves are not in dispute. Random image-level evaluation overstates ADNI1 AUROC by +0.118 on the primary arm (paired-seed interval [+0.096, +0.145] at $n = 5$, descriptive), and the gap survives every pre-specified variation of cohort, backbone, converter rule, acquisition type, input dimensionality and label-linkage strictness (Figure 7; the volumetric arm in §3.3, where the gap is +0.235 against +0.118 in 2D, under a different training recipe).

Three controlled interventions locate the mechanism. Permuting diagnosis labels across participants removes the disease signal entirely and leaves the random split at 0.878 AUROC while both patient-aware protocols sit at chance, which is the shortcut being taken rather than inferred. Substituting contaminated scans into the test partition gives +0.125 AUROC per unit nominal overlap, descriptive of that procedure. Deleting subject identifiers at controlled rates shows how much leakage each protocol admits as provenance decays, with 49% of it prevented by the leakage graph at 10% identifier loss and subject-wise splitting returning 36% of test participants to training at that same level. A widely used public benchmark shows the same failure without any intervention: it is OASIS-1 redistributed under an identifier that merges and splits participants, and a split that respects that identifier still leaves 70% of test images sharing a participant with training, a contamination nothing inside the release can reveal. Its AUROC price, +0.027 on five seeds, is not resolved at that seed count. A linear probe confirms that patient identity is decodable far above chance, across sessions as well as within them, but that quantity is unchanged between leaky and honest protocols and therefore establishes decodability rather than causation; the causal weight rests on the three interventions.

The field-level implication is that AD MRI benchmark results should be read as claims about an evaluation protocol, not only about a model architecture. The principal limitation is single-cohort headline scope: ADNI is the only cohort on which the cost-of-leakage, subgroup and dose-response analyses run, and external-cohort replication (AIBL, UK Biobank) is the immediate next step. SplitGuard-AD provides the reusable infrastructure to run it: frozen split manifests, an executable pre-training GO/NO-GO audit following Algorithm 1, and a benchmark report card.

Acknowledgements

Every training run reported here was executed in one code state (approximately 14.8 GPU-hours in total). The software environment of the node that ran the final arms is released in full; the equivalent records for the earlier nodes were not retained. Most arms ran on a single cloud-hosted NVIDIA H100 instance; the 50-run dose-response matrix was rerun in that same code state on an NVIDIA RTX PRO 4500 instance, the 15-run volumetric arm on an NVIDIA RTX 2000 Ada instance, and the 15-run exact-label-linkage arm together with the 120-run structured-corruption matrix, with the 15-run label-permutation control, on an NVIDIA RTX 4090 instance, so those arms and the ones they are compared with differ in hardware but not in code. We report the slope as a descriptive calibration constant and not as a transferable coefficient, which is how it is read throughout. Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (https://adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. OASIS-1 data were provided in part by OASIS Cross-Sectional Principal Investigators: D. Marcus, R. Buckner, J. Csernansky, J. Morris (P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, P20 MH071616, U24 RR021382).

Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate

This study used retrospective, de-identified secondary imaging data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-1), including a public Kaggle benchmark that we show to be a redistribution of OASIS-1 images. No new human participants were recruited, no new imaging was acquired, and no direct participant contact occurred. ADNI and OASIS obtained ethics approval and informed consent at the original data-collection sites; the present secondary analysis was conducted under their respective data-use agreements and public-access terms. The Kaggle benchmark consists of already-public OASIS-1 slice images, and its provenance recovery was performed against OASIS-1 data under the same terms; the recovered image-to-participant mapping is not redistributed. Because this analysis is limited to secondary de-identified data, institutional review board (IRB) approval was not separately sought at Istanbul Medipol University in accordance with local policy on secondary de-identified data reuse. Participant-level

manifests remain local and are hashed via PBKDF2 with a per-release salt so as not to support cross-dataset participant re-identification; only aggregate result tables are redistributed.

Consent for publication

No individual participant data, images, or identifying details are reproduced in this manuscript or in any released artefact, so participant consent for publication is not applicable. All released tables are aggregate, and participant identifiers in the Tier-3 manifests are replaced by a PBKDF2 derivation under a secret salt.

Use of generative AI

Large-language-model coding assistants were used to implement and revise the released software, including the training, audit, split-generation and analysis scripts and the figure and table generators, and to draft manuscript prose. They were also used to locate defects, two of which are reported in the text because they are the kind of silent, self-consistent error this paper concerns. The study design, the protocol definitions, the choice of cohorts and arms and the interpretation of every result are the authors'. No reported number was produced or altered by an LLM: every quantity is computed by the released scripts from the released manifests, and `scripts/verify_paper_numbers.py` re-checks each against the artefact that produced it, so a value introduced into the text would fail that check rather than survive it. The authors reviewed and edited all such output and take full responsibility for the content of this publication.

Author contributions

Mehmet Berke Isler: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Irem Ilter:** Investigation (OASIS-1 cohort), Formal analysis, Writing – review & editing. **Mehmet Kemal Ozdemir:** Supervision, Methodology, Writing – review & editing, Project administration.

Data and code availability

The SplitGuard-AD framework, the training and analysis pipeline, and the manuscript sources are released at `https://github.com/mehmetberkeisler/splitguard-ad` under the Apache-2.0 licence.

The release boundary is aggregate-only: scripts, paper sources, figures, aggregate result tables and rebuild instructions are public, while raw data, model weights, local manifests, detailed audit rows and per-image predictions remain local. The manifest release boundary is tier-dependent, and the artefacts for each tier are materialised by `scripts/build_release_artefacts.py` into the tracked `release/` directory, with a `RELEASE_MANIFEST.json` recording the row counts and the columns withheld per tier. **Tier 1 (redistributed public benchmark)**: the release includes the frozen filename-key split manifest at image-path level (6,400 rows), its leakage-graph component table (6,400 rows) and the near-duplicate candidate list (1,582 pairs), which together document the split whose contamination §2.3.1 measures. The participant-level mapping to OASIS-1 is not released; `scripts/audit_tier1_ground_truth.py` and `scripts/tier1_split_rules_vs_truth.py` regenerate it, the recovered-participant splits and every

Tier-1 contamination and calibration number from a user's own Kaggle and OASIS-1 downloads. **Tier 2 (OASIS-1)**: image-level identifiers are not ours to redistribute but OASIS-1 access is open to any registered user, so the release includes the frozen split manifest (1,175 rows) and component table (2,180 rows) indexed by the public OASIS subject, session and volume codes only, with no filesystem paths and no clinical fields (CDR, MMSE, education, SES, age, sex), which a third party re-derives from their own download. The projection is enforced in code rather than by convention: the release builder refuses to emit any column on its per-tier restricted list. **Tier 3 (ADNI1)**: under the LONI/ADNI data-use agreement, participant-level identifiers cannot be redistributed. For this tier the release includes *hashed* component-safe manifests, one per frozen split of the five-seed primary arm (220 rows each, one row per single-subject component): each row carries a PBKDF2-HMAC-SHA256 derivation of the subject identifier (600,000 iterations under a secret per-release salt), an integer component-size summary, the binary CN/AD label, and the non-identifying acquisition fields (`scanner_field_strength`, modality flag), but no participant-level identifiers, dates, or image paths. The converter-inclusive arm is not released in this form, because a converting participant contributes both CN and AD scans to one component and the aggregation refuses a component whose label is not uniform rather than choosing one for it. Key stretching rather than a plain digest is required here: ADNI PTIDs are drawn from a space of roughly 10^7 candidates, so a single-round hash of one is invertible by enumeration in under a second, and the salt cannot be published for the same reason. The exact-linkage arm's manifests and the 120 corrupted manifests of §3.9 are withheld under the same agreement, and need not be released: both are deterministic functions of the primary manifest, regenerated by `make_adni_splitguard_split.py --label-confidence clinical_table` and by `run_provenance_stress_test.py --emit-splits` from a reader's own ADNI download. A downstream auditor can therefore verify component sizes, class balance, and cross-partition contamination on Tier 3 without obtaining the underlying ADNI data, and without the release itself re-identifying participants. Data access for the three tiers: Tier 1 is the public Kaggle four-class Alzheimer's MRI dataset [40], itself derived from OASIS-1; Tier 2 (OASIS-1) [41] is available at https://www.oasis-brains.org/; Tier 3 (ADNI1: Complete 3Yr 1.5T) is accessed through the LONI Image and Data Archive (IDA) at https://ida.loni.usc.edu/ under an approved Data Use Agreement. ADNI participant-level study files and image volumes are retained locally and are not redistributed; only the hashed manifests, aggregate result tables, and rebuild scripts are released for that tier.

Reproducibility boundary of the reported runs

Every AUROC in this manuscript comes from `scripts/gpu_program.py` in a single code state, on the GPUs named in the Acknowledgements, with seeded samplers, per-worker augmentation seeding and deterministic cuDNN kernels. The driver, CUDA version and full package list of the final node are released as `runs_gpu/gpu_environment.txt`, frozen by `scripts/gpu_setup.sh` before training rather than reconstructed afterwards, and `requirements.txt` is read off that record. Each run persists its per-image test predictions. `scripts/gpu_postprocess.py` derives the headline AUROCs, intervals, controls and result tables from those predictions and writes them into generated files the manuscript reads, so those numbers are never transcribed by hand; the remaining quantities are computed by the analysis scripts and held to their artefacts by `scripts/verify_paper_numbers.py`, which

fails on any divergence. Those per-image predictions carry ADNI participant identifiers and therefore stay local; what is released is the code that produces them, the aggregate outputs, and `scripts/verify_paper_numbers.py`, which re-derives every numerical claim from those outputs and fails the build on any divergence. Model weights are not released, because weights trained on participant-level data fall under the same data-use terms as the data. Re-running training on different hardware reproduces the protocol ordering, the sign and the approximate magnitude of every effect, but not the third decimal place, and we can state that from measurement rather than expectation: regenerating every arm moved 36 of the 116 compared runs by at least 0.010 AUROC and one of them by 0.084, while every protocol ordering, every gap sign and every interval-level conclusion held. The run-by-run comparison is released as `reports/rerun_comparison.md`, and because each downstream artefact is regenerated from the predictions rather than edited, the manuscript follows those movements automatically.

Supplementary Material

Supplementary Material accompanying this submission contains: (S1) consolidated training hyperparameters for all tiers; (S2) the ten-row ADNI operating-point sensitivity table; (S3) SplitGuard-AD outputs mapped to the CLAIM [24], TRIPOD+AI [26] and PROBAST domain 4 [27] reporting domains; (S4) the dose-response mixed-effects model specification and cluster-robust inference; (S5) the ADNI 180-day linkage-integrity artefact schema; and (S6) the sex-stratified ADNI analysis.

References

[1] World Health Organization. Dementia. WHO; 2023. Available from: https://www.who.int/news-room/fact-sheets/detail/dementia.

[2] Jack CR, Albert MS, Knopman DS, McKhann GM, Sperling RA, Carrillo MC, et al. Introduction to the recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. Alzheimer's & Dementia. 2011;7(3):257–262. https://doi.org/10.1016/j.jalz.2011.03.004.

[3] Korolev S, Safiullin A, Belyaev M, Dodonova Y. Residual and plain convolutional neural networks for 3D brain MRI classification. In: 2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI); 2017. p. 835–838.

[4] Lian C, Liu M, Zhang J, Shen D. Hierarchical Fully Convolutional Network for Joint Atrophy Localization and Alzheimer's Disease Diagnosis Using Structural MRI. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2020;42(4):880–893. https://doi.org/10.1109/TPAMI.2018.2889096.

[5] Qiu S, Joshi PS, Miller MI, Xue C, Zhou X, Karjadi C, et al. Development and validation of an interpretable deep learning framework for Alzheimer's disease classification. Brain. 2020;143(6):1920–1933. https://doi.org/10.1093/brain/awaa137.

[6] Young VM, Gates S, Garcia LY, Salardini A. Data Leakage in Deep Learning for Alzheimer's Disease Diagnosis: A Scoping Review of Methodological Rigor and Performance Inflation. Diagnostics. 2025;15(18):2348. https://doi.org/10.3390/diagnostics15182348.

[7] Wen J, Thibeau-Sutre E, Diaz-Melo M, Samper-González J, Routier A, Bottani S, et al. Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation. Medical Image Analysis. 2020;63:101694. https://doi.org/10.1016/j.media.2020.101694.

[8] Yagis E, García Seco de Herrera A, Citi L. Effect of data leakage in brain MRI classification using 2D convolutional neural networks. Scientific Reports. 2021;11(1):22544. https://doi.org/10.1038/s41598-021-01681-w.

[9] Rosenblatt M, Tejavibulya L, Jiang R, Noble S, Scheinost D. Data leakage inflates prediction performance in connectome-based machine learning models. Nature Communications. 2024;15(1):1829. https://doi.org/10.1038/s41467-024-46150-w.

[10] Geirhos R, Jacobsen JH, Michaelis C, Zemel R, Brendel W, Bethge M, et al. Shortcut learning in deep neural networks. Nature Machine Intelligence. 2020;2:665–673. https://doi.org/10.1038/s42256-020-00257-z.

[11] Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann EK. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLOS Medicine. 2018;15(11):e1002683. https://doi.org/10.1371/journal.pmed.1002683.

[12] DeGrave AJ, Janizek JD, Lee SI. AI for radiographic COVID-19 detection selects shortcuts over signal. Nature Machine Intelligence. 2021;3:610–619. https://doi.org/10.1038/s42256-021-00338-7.

[13] Oakden-Rayner L, Dunnmon J, Carneiro G, Ré C. Hidden Stratification Causes Clinically Meaningful Failures in Machine Learning for Medical Imaging. In: Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL); 2020. p. 151–159.

[14] Schwarz CG, Kremers WK, Therneau TM, Sharp RR, Gunter JL, Vemuri P, et al. Identification of Anonymous MRI Research Participants with Face-Recognition Software. New England Journal of Medicine. 2019;381(17):1684–1686. https://doi.org/10.1056/NEJMc1908881.

[15] Wachinger C, Golland P, Kremen W, Fischl B, Reuter M. BrainPrint: A discriminative characterization of brain morphology. NeuroImage. 2015;109:232–248. https://doi.org/10.1016/j.neuroimage.2015.01.032.

[16] Puglisi L, Eshaghi A, Parker GJM, Barkhof F, Alexander DC, Ravi D. DeepBrainPrint: A novel contrastive framework for brain MRI re-identification. Proceedings of Machine Learning Research, Medical Imaging with Deep Learning. 2023;.

[17] Liem F, Varoquaux G, Kynast J, Beyer F, Kharabian Masouleh S, Huntenburg JM, et al. Predicting brain-age from multimodal imaging data captures cognitive impairment. NeuroImage. 2017;148:179–188. https://doi.org/10.1016/j.neuroimage.2016.11.005.

[18] Roberts M, Driggs D, Thorpe M, Gilbey J, Yeung M, Ursprung S, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. Nature Machine Intelligence. 2021;3:199–217. https://doi.org/10.1038/s42256-021-00307-0.

[19] Varoquaux G, Cheplygina V. Machine learning for medical imaging: methodological failures and recommendations for the future. npj Digital Medicine. 2022;5:48. https://doi.org/10.1038/s41746-022-00592-y.

[20] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 2023;4(9):100804. https://doi.org/10.1016/j.patter.2023.100804.

[21] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.

[22] Cardoso MJ, et al.: MONAI: An open-source framework for deep learning in healthcare. arXiv preprint arXiv:2211.02701.

[23] Joeres R, Blumenthal DB, Kalinina OV. Data splitting to avoid information leakage with DataSAIL. Nature Communications. 2025;16(3337). https://doi.org/10.1038/s41467-025-58606-8.

[24] Tejani AS, Klontzas ME, Gatti AA, Mongan JT, Moy L, Park SH, et al. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update. Radiology: Artificial Intelligence. 2024;6(4):e240300. https://doi.org/10.1148/ryai.240300.

[25] Mongan J, Moy L, Kahn CE Jr. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): A Guide for Authors and Reviewers. Radiology: Artificial Intelligence. 2020;2(2):e200029. https://doi.org/10.1148/ryai.2020200029.

[26] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning. BMJ. 2024;385:e078378. https://doi.org/10.1136/bmj-2023-078378.

[27] Wolff RF, Moons KGM, Riley RD, Whiting PF, Westwood M, Collins GS, et al. PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies. Annals of Internal Medicine. 2019;170(1):51–58. https://doi.org/10.7326/M18-1376.

[28] Chaibub Neto E, Pratap A, Perumal TM, Tummalacherla M, Bot BM, Trister AD, et al.: Learning Disease vs Participant Signatures: a permutation test approach to detect identity confounding in machine learning diagnostic applications. Available from: https://arxiv.org/abs/1712.03120.

[29] Rumala DJ. How You Split Matters: Data Leakage and Subject Characteristics Studies in Longitudinal Brain MRI Analysis. In: Clinical Image-Based Procedures, Fairness of AI in Medical Imaging, and Ethical and Philosophical Issues in Medical Imaging (FAIMI 2023). vol. 14242 of Lecture Notes in Computer Science. Springer Nature Switzerland; 2023. p. 235–245.

[30] Ghiasi MM, Falck RS, Liu-Ambrose T, Tam RC. Explainable machine learning for predicting longitudinal dementia status: Establishing a leakage-free benchmark. PLOS Digital Health. 2026;5(5):e0001409. https://doi.org/10.1371/journal.pdig.0001409.

[31] Sifat M, Akter S, Islam A, Hamid ME, Miah ASM, Hassan N, et al.: 3D MRI-Based Alzheimer's Disease Classification Using Multi-Modal 3D CNN with Leakage-Aware Subject-Level Evaluation. Audits a Kaggle redistribution of OASIS-1 supplied as pre-extracted 2D JPEG slices, tracing subject identifiers from slice captions. Available from: https://arxiv.org/abs/2603.17304.

[32] Vangala BP, Guda S, Peddi L, Vangala N.: Scores That Hold, Benchmarks That Leak: Measuring Dataset Contamination in Public Brain-Tumor MRI Classification. Recovers

patient identifiers for 12.6% of a public brain-tumor test split by matching to the figshare originals; reports that removing near-duplicate test images leaves balanced accuracy unchanged, without retraining. Available from: https://arxiv.org/abs/2610.00421.

[33] Ellis KA, Bush AI, Darby D, De Fazio D, Foster J, Hudson P, et al. The Australian Imaging, Biomarkers and Lifestyle (AIBL) study of aging: methodology and baseline characteristics of 1112 individuals recruited for a longitudinal study of Alzheimer's disease. International Psychogeriatrics. 2009;21(4):672–687. https://doi.org/10.1017/S1041610209009405.

[34] Sudlow C, Gallacher J, Allen N, Beral V, Burton P, Danesh J, et al. UK Biobank: An Open Access Resource for Identifying the Causes of a Wide Range of Complex Diseases of Middle and Old Age. PLOS Medicine. 2015;12(3):e1001779. https://doi.org/10.1371/journal.pmed.1001779.

[35] LaMontagne PJ, Benzinger TLS, Morris JC, Keefe S, Hornbeck R, Xiong C, et al. OASIS-3: Longitudinal Neuroimaging, Clinical, and Cognitive Dataset for Normal Aging and Alzheimer Disease. medRxiv. 2019;https://doi.org/10.1101/2019.12.13.19014902.

[36] Glocker B, Robinson R, Castro DC, Dou Q, Konukoglu E.: Machine Learning with Multi-Site Imaging Data: An Empirical Study on the Impact of Scanner Effects. Medical Imaging meets NeurIPS Workshop; arXiv:1910.04597.

[37] Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics. 2007;8(1):118–127. https://doi.org/10.1093/biostatistics/kxj037.

[38] Fortin JP, Parker D, Tunç B, Watanabe T, Elliott MA, Ruparel K, et al. Harmonization of multi-site diffusion tensor imaging data. NeuroImage. 2017;161:149–170. https://doi.org/10.1016/j.neuroimage.2017.08.047.

[39] Pomponio R, Erus G, Habes M, Doshi J, Srinivasan D, Mamourian E, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. NeuroImage. 2020;208:116450. https://doi.org/10.1016/j.neuroimage.2019.116450.

[40] : Alzheimer's Dataset (4 class of images). 6,400 2D brain-MRI JPEG images in four severity folders, uniform 176 × 208 pixels, released without subject identifiers, acquisition metadata or a statement of origin. Shown in this work by content matching to be 32 axial slices each of 200 OASIS-1 participants, with folder labels equal to their CDR. Kaggle public dataset. Available from: https://www.kaggle.com/datasets/tourist55/alzheimers-dataset-4-class-of-images.

[41] Marcus DS, Wang TH, Parker J, Csernansky JG, Morris JC, Buckner RL. Open Access Series of Imaging Studies (OASIS): Cross-sectional MRI Data in Young, Middle Aged, Nondemented, and Demented Older Adults. Journal of Cognitive Neuroscience. 2007;19(9):1498–1507. https://doi.org/10.1162/jocn.2007.19.9.1498.

[42] Weiner MW, Veitch DP, Aisen PS, Beckett LA, Cairns NJ, Green RC, et al. The Alzheimer's Disease Neuroimaging Initiative: A review of papers published since its inception. Alzheimer's & Dementia. 2012;8(1 Suppl):S1–S68. https://doi.org/10.1016/j.jalz.2011.09.172.

[43] Jack CR, Bernstein MA, Fox NC, Thompson P, Alexander G, Harvey D, et al. The Alzheimer's Disease Neuroimaging Initiative (ADNI): MRI methods. Journal of Magnetic Resonance Imaging. 2008;27(4):685–691. https://doi.org/10.1002/jmri.21049.

[44] Petersen RC, Aisen PS, Beckett LA, Donohue MC, Gamst AC, Harvey DJ, et al. Alzheimer's Disease Neuroimaging Initiative (ADNI): Clinical characterization. Neurology. 2010;74(3):201–209. https://doi.org/10.1212/WNL.0b013e3181cb3e25.

[45] Efron B, Tibshirani R. Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. Statistical Science. 1986;1(1):54–75. https://doi.org/10.1214/ss/1177013815.

[46] Carpenter J, Bithell J. Bootstrap confidence intervals: when, which, what? A practical guide for medical statisticians. Statistics in Medicine. 2000;19(9):1141–1164. https://doi.org/10.1002/(SICI)1097-0258(20000515)19:9<1141::AID-SIM479>3.0.CO;2-F.

[47] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR); 2016. p. 770–778.

[48] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. In: Proceedings of the 22nd International Conference on Machine Learning (ICML); 2005. p. 625–632.

[49] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML). vol. 70 of Proceedings of Machine Learning Research; 2017. p. 1321–1330. Available from: https://proceedings.mlr.press/v70/guo17a.html.

[50] Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the Achilles heel of predictive analytics. BMC Medicine. 2019;17(1):230. https://doi.org/10.1186/s12916-019-1466-7.

[51] Bergsma W. A bias-correction for Cramér's V and Tschuprow's T. Journal of the Korean Statistical Society. 2013;42(3):323–328. https://doi.org/10.1016/j.jkss.2012.10.002.

[52] Bashyam VM, Doshi J, Erus G, Srinivasan D, Abdulkadir A, Singh A, et al. Deep Generative Medical Image Harmonization for Improving Cross-Site Generalization in Deep Learning Predictors. Journal of Magnetic Resonance Imaging. 2022;55(3):908–916. https://doi.org/10.1002/jmri.27908.

[53] Rouzrokh P, Khosravi B, Faghani S, Moassefi M, Vera Vázquez D, Singh Y, et al. Mitigating Bias in Radiology Machine Learning: 1. Data Handling. Radiology: Artificial Intelligence. 2022;4(5):e210290. https://doi.org/10.1148/ryai.210290.

[54] Hua X, Hibar DP, Lee S, Toga AW, Jack CR Jr, Weiner MW, et al. Sex and age differences in atrophic rates: an ADNI study with $n = 1368$ MRI scans. Neurobiology of Aging. 2010;31(8):1463–1480. https://doi.org/10.1016/j.neurobiolaging.2010.04.033.

[55] Sundermann EE, Maki PM, Rubin LH, Lipton RB, Landau S, Biegon A. Female advantage in verbal memory: Evidence of sex-specific cognitive reserve. Neurology. 2016;87(18):1916–1924. https://doi.org/10.1212/WNL.0000000000003288.

[56] Mielke MM, Vemuri P, Rocca WA. Clinical epidemiology of Alzheimer's disease: assessing sex and gender differences. Clinical Epidemiology. 2014;6:37–48. https://doi.org/10.2147/CLEP.S37929.

[57] Rajan KB, Weuve J, Barnes LL, McAninch EA, Wilson RS, Evans DA. Population estimate of people with clinical Alzheimer's disease and mild cognitive impairment in the United States (2020–2060). Alzheimer's & Dementia. 2021;17(12):1966–1975. https://doi.org/10.1002/alz.12362.

[58] Thomas G, Smith EE, Ismail Z, Kirkham JG, Roach P, Goodarzi Z, et al. Potential Eligibility for Disease Modifying Therapies Among Individuals Referred to A Tertiary Care Memory Clinic in Calgary, Alberta, Canada: Implications for AD Biomarker Testing (Prospective Registry for Persons with Memory Symptoms, PROMPT). Alzheimer's & Dementia. 2024;https://doi.org/10.1002/alz.091118.

[59] Jack CR, Bennett DA, Blennow K, Carrillo MC, Dunn B, Haeberlein SB, et al. NIA-AA Research Framework: Toward a biological definition of Alzheimer's disease. Alzheimer's & Dementia. 2018;14(4):535–562. https://doi.org/10.1016/j.jalz.2018.02.018.

[60] Lin JS, O'Connor E, Rossom RC, Perdue LA, Eckstrom E. Screening for cognitive impairment in older adults: A systematic review for the U.S. Preventive Services Task Force. Annals of Internal Medicine. 2013;159(9):601–612. https://doi.org/10.7326/0003-4819-159-9-201311050-00730.

[61] Wittenberg R, Knapp M, Hu B, Comas-Herrera A, King D, Rehill A, et al. The costs of dementia in England. International Journal of Geriatric Psychiatry. 2019;34(7):1095–1103. https://doi.org/10.1002/gps.5113.